



UNIL | Université de Lausanne

Faculté de biologie
et de médecine

Ecole de biologie

**RESOURCES VARIABILITY AS A DRIVER OF COOPERATIVE
EXCHANGE AND REDUCED CONFLICT**

**Travail de Maîtrise universitaire ès Sciences en comportement, évolution et
conservation, spécialisation « écologie et évolution computationnelles »**
*Master Thesis of Science in Behaviour, Evolution and Conservation,
specialisation « Computational Ecology and Evolution »*

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09 (janvier, 2026)

Contents

1	Introduction	6
2	Model	10
2.1	Actions and interactions	10
2.1.1	Pairwise interactions	11
2.2	Traits and decision process	13
2.3	Analysis	14
2.3.1	Fitness definition	14
2.3.2	Environmental variability	15
2.3.3	Initial conditions	16
2.3.4	Simulation comparisons	17
2.4	Materials	17
3	Results	18
3.1	Low-variability environment	18
3.2	High-variability environments	19
3.2.1	Uncorrelated variability environment	19
3.2.2	Rich-poor variability environment	21
3.2.3	Trade-off variability environment	24
3.2.4	Avoiders cannot escape conflict	26
3.2.5	Complementary resources	28
4	Discussion	29
5	Acknowledgements	34
S1	Supplementary Material	39
S1.1	Trade formulation	39
S1.2	Avoidance proportions	42
S1.2.1	Constrained avoidance	43

S1.2.2 Complementary resources	44
S1.3 Within population variance	44
S1.3.1 Low variability	46
S1.3.2 uncorrelated variability	48
S1.3.3 Rich–poor variability	50
S1.3.4 Trade–off variability	52
S1.4 Constrained-avoidance case	53
S1.5 Complementary resources	58

List of Figures

1 Schematic representation of the decision process	14
2 Joint distributions of initial resource endowments	16
3 Time series of mean belligerence b and mean dissatisfaction threshold λ .	19
4 Action proportions among non-avoiding agents in low- and uncorrelated- variability environments across costs	21
5 Action proportions among non-avoiding agents in low-variability and rich–poor environments across costs	23
6 Time series of mean belligerence b and mean dissatisfaction threshold λ .	24
7 Action proportions among non-avoiding agents in the low-variability and trade-off environments across scenario	26
S1 Action versus avoidance proportions across environmental variability sce- narios	42
S2 Action versus avoidance proportions across environmental variability sce- narios when avoidance is constrained	43
S3 Action versus avoidance proportions across environmental variability sce- narios under the Cobb–douglas formulation	44
S4 Trait dynamics and equilibrium distributions under the low-variability sce- nario and low conflict costs	46

S5	Trait dynamics and equilibrium distributions under the low-variability scenario and high conflict costs	47
S6	Trait dynamics and equilibrium distributions under the uncorrelated variability scenario and low conflict costs	48
S7	Trait dynamics and equilibrium distributions under the uncorrelated variability scenario and high conflict costs	49
S8	Trait dynamics and equilibrium distributions under the rich–poor variability scenario and low conflict costs	50
S9	Trait dynamics and equilibrium distributions under the rich–poor variability scenario and high conflict costs	51
S10	Trait dynamics and equilibrium distributions under the trade–off variability scenario and low conflict costs	52
S11	Trait dynamics and equilibrium distributions under the trade–off variability scenario and high conflict costs	53
S12	Time series of mean belligerence b and dissatisfaction threshold λ under low- and uncorrelated-variability environments in the constrained-avoidance case	54
S13	Time series of mean belligerence b and dissatisfaction threshold λ under the trade-off and rich–poor variability environments in the constrained-avoidance case	55
S14	Action proportions among agents in low-variability and uncorrelated environments across costs in the constrained-avoidance case	56
S15	Action proportions among agents in low-variability and rich–poor environments across costs in the constrained-avoidance case	57
S16	Action proportions among agents in low-variability and trade–off environments across costs in the constrained-avoidance case	58
S17	Time series of mean belligerence b and dissatisfaction threshold λ under low- and uncorrelated-variability environments in the Cobb–Douglas fecundity case	59

S18	Time series of mean belligerence b and dissatisfaction threshold λ under the trade-off and rich-poor variability environments in the Cobb-Douglas fecundity case	60
S19	Action proportions among non-avoiding agents in the low-variability and uncorrelated environments across costs under Cobb-Douglas fecundity case	61
S20	Action proportions among non-avoiding agents in the low-variability and rich-poor environments across costs under Cobb-Douglas fecundity case	62
S21	Action proportions among non-avoiding agents in the low-variability and trade-off environments across costs under Cobb-Douglas fecundity case .	63

List of Tables

1	Description of the different action choices available to agents.	11
2	Summary of the possible interactions depending on the action choices of paired agents.	12
3	Summary of the model parameters used in the analysis.	17

Abstract

Social interactions over limited resources can generate both conflict and cooperation, yet these interactions are often studied in isolation and with limited attention to environmental variability. Here, we identify drivers through which resource variability can reduce violent conflict and promote cooperative exchange. We designed an agent-based model in which agents acquire two resources and then decide whether to avoid interaction or engage through conflict or exchange. Agents possess a dissatisfaction threshold that determines whether they engage in interaction based on their current resource state. When interaction occurs, a belligerence trait governs the choice between conflict and cooperative exchange.

We simulate populations under different forms of resource variability. When variability is high, more agents experience resource shortfalls, making engagement more likely and concentrating interactions among the poorest agents, for whom the gains from conflict are limited. As a result, cooperative exchange dominates among interacting agents, while war collapses to near-zero frequencies. This effect is strongest under extreme variability, such as rich–poor environments. Finally, our results show how environmental variability can change how agents interact, offering a simple explanation for the coexistence of conflict and cooperation.

Les interactions sociales lorsqu’il y a des ressources limitées peuvent engendrer à la fois des conflits et de la coopération, mais ces interactions sont souvent étudiées de manière isolée et avec une attention limitée à l’impact de la variabilité environnementale. Ici, nous identifions des facteurs simples via lesquels la variabilité environnementale peut réduire les conflits et favoriser les échanges coopératifs. Nous avons fait un modèle évolutif basé sur des agents, dans lequel les agents acquièrent deux ressources puis décident soit d’éviter l’interaction, soit de s’engager par le conflit ou l’échange. Les agents possèdent un seuil d’insatisfaction qui détermine s’ils agissent ou pas, en fonction de leurs ressources la plus limitante. Lorsqu’ils décident d’agir, un trait, la belligérance détermine s’ils choisissent le conflit ou la coopération.

Nous simulons des populations sous différentes formes d’inégalités de ressources. Lorsque la variabilité est élevée, davantage d’agents subissent des pénuries, ce qui

rend l'engagement plus probable tout en concentrant les interactions parmi les agents les plus pauvres, pour lesquels les gains issus du conflit sont limités. En conséquence, l'échange coopératif est l'interaction qui domine parmi nos agents, tandis que la guerre s'effondre à des fréquences proches de zéro car moins stratégiquement viable. Cet effet est le plus marqué sous une variabilité extrême, comme dans des environnements riches-pauvres. Ainsi, nos résultats montrent comment la variabilité environnementale peut modifier la manière dont les agents interagissent entre eux, offrant une explication simple à la coexistence entre conflit et coopération.

1 Introduction

Social behaviour can be defined as any behaviour expressed by an organism that affects the fitness of other conspecifics, as well as its own (Krebs et al., 1982). Social behaviours are widespread across the tree of life and arise whenever organisms interact over finite resources such as food, territory or mating opportunities (Frank, 2007). Organisms thus face recurrent decisions about how to respond to competition, with possible strategies ranging from avoidance to conflictual appropriation or cooperative exchange.

Both conflictual and cooperative social behaviours are observed across a wide range of organisms. In viruses, the presence of other viral particles within the same host can increase individual fitness by improving the ability to overcome host immune defences, while excessive viral density leads to competition for host resources, generating conflictual interactions among conspecific viral particles (Sanjuán, 2021). In bacteria, social behaviour includes cooperative interactions such as biofilm formation, which increases resistance to environmental stress, as well as competitive interactions over space and nutrients (Li and Tian, 2012).

In multicellular organisms, conflictual social behaviours often take the form of aggression or warfare over resources. For example, wolves (*Canis lupus*) engage in coordinated aggression over territory and prey, while cooperatively breeding birds such as the greater ani (*Crotophaga major*) form coalitions to raid neighbouring nests (Metz et al., 2011; Almstead et al., 2021). Cooperative social behaviours are likewise common and include

coordinated foraging, collective defence, and resource exchange.

Eusocial insects provide some of the clearest examples of the coexistence of conflict and cooperation. Ants and termites engage in large-scale warfare against rival colonies, while also forming extensive networks of interconnected nests that exchange resources and maintain persistent cooperative routes (Hölldobler and Wilson, 2009).

Among primates, social behaviour similarly spans a broad spectrum. Some species can differ markedly in their dominant social strategies: chimpanzees are known for coordinated territorial aggression, whereas bonobos exhibit comparatively tolerant and affiliative interactions, including food sharing (Wrangham and Glowacki, 2012; Surbeck and Hohmann, 2015). Humans represent the most extensively documented example within primates. Human societies display a wide range of social behaviours, from organised collective violence to elaborate forms of cooperation such as trade, alliances, and cultural exchange (Bowles, 2009; Rusch, 2022). This duality has been described as the “paradox of peace and war” and has fuelled long-standing debates in evolutionary anthropology concerning the prevalence of conflict and cooperation in early human societies (Fry, 2013, 2025; Glowacki, 2024; Wilson, 2025). These examples illustrate that conflictual and cooperative social behaviours represent alternative responses to competition over limited resources.

When considering competition, a common expectation is that resource scarcity or unpredictability may promote risk-taking and, consequently, conflictual strategies. When individuals face the prospect of falling below subsistence levels, engaging in risky actions such as aggression or appropriation may offer a chance to secure otherwise unattainable gains. Recent theoretical and empirical work has explored this idea by explicitly considering deprivation and inequality, suggesting that low resource levels or unequal distributions can increase the attractiveness of risky or violent behaviours, particularly when individuals have little to lose (Smirnov et al., 2007; De Courson et al., 2023, 2025). However, while the relationship between scarcity, inequality, and conflict is appealing, conclusive empirical support remains limited and context-dependent (Abbink et al., 2018).

At the same time, resource scarcity and environmental variability may also favour avoidance, tolerance, or cooperative strategies rather than escalation into costly conflicts.

Risk aversion, deterrence, and the high costs of aggression can lead individuals or groups to refrain from conflict even under adverse conditions. For example, incorporating risk aversion and deterrence into models of intergroup interaction can generate stable patterns of tolerance rather than conflict (Rusch, 2014). Archaeological, experimental, and theoretical evidence further suggests that harsh or variable environments can promote cooperation and resource conservation, although these effects may weaken or reverse when opportunities for conflict are present (Spikins et al., 2021; Safarzynska, 2018, 2017; Inaba and Akiyama, 2024).

Understanding how such divergent responses emerge and persist has attracted sustained interest across multiple disciplines. A large body of theory has examined the evolution of conflictual and cooperative behaviours, using a variety of modelling approaches across economics, evolutionary biology, and the social sciences (Choi and Bowles, 2007; Bowles, 2009; Lehmann and Feldman, 2008; Rusch, 2014; De Dreu et al., 2020; Rusch, 2022; Rodrigues et al., 2023; Böhm et al., 2020). In economics and game theory, models typically analyse equilibrium strategies in which individuals allocate effort between productive activities and conflict behaviours such as fighting or appropriation, treating agents as rational actors who freely choose behaviours to maximise payoffs. In these frameworks, conflict is often interpreted as a subsistence strategy, where resources produced by some agents are appropriated by others.

In parallel, evolutionary game theory and population-genetic models treat social behaviours as genetically or culturally inherited traits and focus on their evolutionary stability rather than individual optimisation. This literature has extensively investigated the evolution of conflictual behaviours such as belligerence or aggressivity, as well as cooperative behaviours such as helping or resource sharing or effort, often within group-structured populations where relatedness, limited dispersal, or demographic stratification play a central role as key mechanisms driving the selection pressure of those traits (Choi and Bowles, 2007; Lehmann and Feldman, 2008; Dyble, 2021; Micheletti et al., 2018; Sankey et al., 2022). Together, these approaches have clarified how costs, benefits, and demographic structure shape the evolution of conflict and cooperation.

However, in these papers, conflict and cooperation are most often studied in isolation, within separate frameworks. Models of conflict typically focus on the conditions favouring aggression or warfare, while models of cooperation focus on mechanisms that stabilise helping or exchange, such as reciprocity, punishment, or repeated interactions (West et al., 2007; Bausch, 2014; Gross et al., 2023). As a result, relatively few models allow conflict and cooperation behaviours to emerge as alternative strategies within the same decision-making process, making it difficult to assess how organisms choose between harming others and engaging in mutually beneficial exchange when facing identical environmental constraints.

Moreover, although environmental conditions are frequently invoked as drivers of social behaviour, explicit treatment of environmental stochasticity remains limited. Resource availability is often assumed to be constant or implicit, particularly in models of the evolution of conflict behaviour, whereas recent work on cooperation suggests that variation in resource availability can strongly influence the stability of cooperative exchanges (Inaba and Akiyama, 2024; Spikins et al., 2021). This limits our understanding of how environmental variability shapes the balance between conflictual and cooperative behaviours. Addressing this gap requires examining how environmental stochasticity influences the evolution of social behaviour when conflict and cooperation strategies are both available responses to competition over finite resources.

In this thesis, we combined key elements from these different strands of research to contribute in two main ways. First, we consider conflictual and cooperative social behaviours within a single decision framework, treating them as alternative responses to competition over finite resources. To do so, we adopt a simplified perspective that focuses on social behaviour as the outcome of decision-making under shared environmental conditions. Second, we examine how this choice between conflict and cooperation is shaped by environmental variability, by explicitly incorporating stochasticity in resource availability. We then analyse how different scenarios of environmental variation influence which social behaviour is favoured when both conflictual and cooperative strategies are available.

2 Model

To investigate how environmental variability shapes the evolution of conflict and cooperation, we developed an agent-based evolutionary model in which agents inhabit environments containing two resources. Before defining each step of the simulation, we first give a brief overview of the life cycle events. At the start of each generation, every agent acquires quantities of two resources, denoted r_1 and r_2 . Resource values are set by the ecological scenario under consideration, ranging from low variability environments to highly heterogeneous conditions such as trade-offs or strong inequality between individuals.

Once resources are assigned, agents are randomly paired and engage in pairwise interactions. During each encounter, agents may (i) avoid interaction, (ii) engage in conflict, or (iii) trade. The interaction outcome determines how resources are redistributed between the two partners, resulting in resource gains, resource losses, or unchanged holdings, depending on the actions taken by both agents.

After all interactions within a generation have occurred, agents reproduce in a well-mixed population. Reproductive success depends on the resources accumulated during interactions, with agents holding more resources contributing more offspring to the next generation. Offspring inherit the behavioural traits of their parents, subject to mutation, and constitute the population of agents in the following generation. Repetition of this life cycle allows behavioural traits affecting avoidance, conflict, and cooperative exchange to evolve under selection across different environmental scenarios.

2.1 Actions and interactions

The population consists of G asexual haploid agents. At each generation, agents receive initial quantities of two resources, r_1 and r_2 , according to the environmental scenario. Agents are then randomly paired, forming $G/2$ independent pairwise interactions.

Each agent independently selects an action $A \in \{a, c, t\}$, corresponding to avoidance, conflict, or trade, respectively; the interpretation of each action is summarised in Table 1.

Action	Interpretation
Avoidance ($A = a$)	Neutral behaviour; the agent avoids interaction and retains its resources.
Conflict ($A = c$)	Conflictual behaviour; the agent attempts to seize resources through conflict.
Trade ($A = t$)	Mutually beneficial behaviour; the agent attempts to exchange resources.

Table 1: Description of the different action choices available to agents.

2.1.1 Pairwise interactions

Let agents i and j form a pair. The combination of their chosen actions determines the outcome of the interaction and the resulting redistribution of resources, as follows (see Table 2 for summary).

Avoidance. If at least one agent chooses avoidance ($A_i = a$ or $A_j = a$), no interaction takes place (We also consider later the case in which an avoiding agent is unable to escape a conflictual agent). The avoiding agent prevents any contact with its partner, thus eliminating the risk of resource loss or conflict. As a result, no resources are transferred and both agents retain their initial resource endowments. Final resource holdings, denoted r'_1 and r'_2 , therefore satisfy:

$$\begin{aligned} r'_{1i} &= r_{1i}, & r'_{2i} &= r_{2i}, \\ r'_{1j} &= r_{1j}, & r'_{2j} &= r_{2j}. \end{aligned}$$

Raid. If $A_i = c$ and $A_j = t$, agent i initiates conflict while agent j does not resist. The non-fighting agent relinquishes its resources, which are seized by the fighting agent. Final resource holdings are therefore:

$$\begin{aligned} r'_{1i} &= r_{1i} + r_{1j}, & r'_{1j} &= 0, \\ r'_{2i} &= r_{2i} + r_{2j}, & r'_{2j} &= 0. \end{aligned}$$

The reverse case applies symmetrically when $a_i = t$ and $a_j = c$.

War ($A_i = A_j = 1$). When both agents choose conflict, a war occurs. One agent is selected at random as the winner, with probability $1/2$. If agent i wins, final resource holdings are given by:

$$\begin{aligned} r'_{1i} &= r_{1i} + r_{1j}, & r'_{1j} &= 0, \\ r'_{2i} &= r_{2i} + r_{2j}, & r'_{2j} &= 0. \end{aligned}$$

The losing agent additionally incurs a fecundity cost C_w , representing physiological costs associated with fighting such as injury. Larger values of C_w correspond to more intense and destructive conflicts.

Trade ($A_i = A_j = 2$). When both agents choose trade, resources are exchanged according to a bargaining solution (Kalai and Smorodinsky, 1975). The bargaining outcome is Pareto-efficient and maximizes equal relative proportional gains in fecundity for both agents. In simulations, this corresponds to selecting the allocation that maximizes fecundity gains subject to the constraint that both agents experience similar proportional increases. The resulting post-trade resource holdings (r'_{1i}, r'_{2i}) and (r'_{1j}, r'_{2j}) are determined by this bargaining solution, which is described in detail in Supplementary S1.1.

As described below, the bargaining outcome depends on the definition of fecundity, and we therefore consider two alternative fecundity formulations (see Section 2.3.1).

Action pair	Interaction	Outcome
(c, c)	War	Winner takes all resources; loser pays cost C_w .
(c, t) or (t, c)	Raid	Fighter seizes all opponent resources.
(t, t)	Trade	Nash bargaining exchange of resources.
(a) and (a, c, t)	Avoidance	No transfer of resources.

Table 2: Summary of the possible interactions depending on the action choices of paired agents.

2.2 Traits and decision process

We study the evolution of two quantitative behavioural traits that jointly influence agents' action choices dependently on their level of resources and interaction outcomes. Both traits are continuous and bounded between 0 and 1.

The first trait, λ_i , is a dissatisfaction threshold capturing the minimum level of resources an agent tolerates before being willing to act rather than avoid interaction. Agents whose resource holdings fall below this threshold engage with others, either through conflict or trading.

The second trait, b_i , represents belligerence and determines the probability that an agent chooses conflict rather than trade when it decides to act.

The influence of these traits on behaviour is implemented through a two-step decision process.

Step 1: Avoidance versus engagement Each agent i compares its limiting resource level

$$\rho_i = \min(r_{1i}, r_{2i})$$

to its dissatisfaction threshold λ_i . If $\rho_i \geq \lambda_i$, the agent avoids interaction and chooses action a (avoidance). If $\rho_i < \lambda_i$, the agent engages in social interaction, either conflictual or cooperative.

Step 2: Choice of action conditional on engagement Conditional on engagement, the agent chooses between conflict and trade according to its belligerence trait b_i . Specifically,

$$\Pr(a_i = c) = b_i, \quad \Pr(a_i = t) = 1 - b_i.$$

The decision process is summarised schematically in Figure 1.

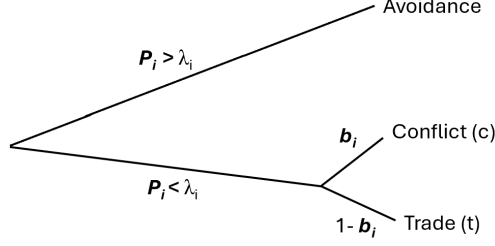


Figure 1: Schematic representation of the decision process leading to avoidance, conflict (c), or trade (t), conditional on the comparison between the limiting resource level ρ_i and the dissatisfaction threshold λ_i , and on the belligerence trait b_i of agent i .

2.3 Analysis

We analyse the joint evolution of the dissatisfaction threshold λ and the belligerence trait b across contrasting environmental scenarios, and compare the resulting evolutionary outcomes.

2.3.1 Fitness definition

Post-interaction resource holdings determine fecundity. Specifically, the payoff (r'_{1i}, r'_{2i}) obtained after interactions is translated into fecundity according to Liebig's law of the minimum (Liebig, 1840):

$$w_i = \min(r'_{1i}, r'_{2i}).$$

Under this assumption, fecundity is determined by the scarcest resource, and the two resources are not complementary. This formulation represents situations in which agent reproduction is limited by a single critical resource and serves as the baseline fitness definition for the analysis, with fecundity determining the number of offspring produced by agent i .

In addition to this baseline case, we also consider a scenario in which fecundity depends on the joint availability of both resources and is defined as the geometric mean:

$$w_i = \sqrt{r'_{1i} r'_{2i}}.$$

This corresponds to a Cobb–Douglas function with equal exponents, a standard formulation in economics (Cobb and Douglas, 1928), and captures situations in which the two resources are complementary. Under this definition, balanced access to both resources yields higher fecundity than the accumulation of a single resource alone.

2.3.2 Environmental variability

At the beginning of each generation, agents receive their initial resource endowments according to one of four environmental scenarios, each capturing a distinct form of environmental or social constraint (Figure 2).

1. **Low-variability environment.** Resources for each agent are drawn independently from a truncated normal distribution with mean 0.5 and standard deviation 0.1, bounded between 0 and 1. This scenario represents a stable and relatively homogeneous environment with limited inequality and serves as a baseline for comparison.
2. **Uncorrelated variability.** Resources for each agent are drawn independently from a uniform distribution over the interval $[0, 1]$. This scenario captures highly stochastic environments in which resource availability varies widely across agents, without systematic correlation between resource types.
3. **Rich–poor environment.** The population is divided into two wealth categories. Half of the population consists of agents in the poor category, for whom both resources are drawn from a low-valued Beta distribution. The remaining half consists of agents in the rich category, for whom both resources are drawn from a high-valued Beta distribution. This scenario represents strong between-agent asymmetries in resource access, which may arise from social inequality or from asymmetric access to resources.
4. **Trade-off environment.** Agents are rich in one resource and poor in the other. For each agent, one resource is drawn from a high-valued Beta distribution and the other from a low-valued Beta distribution. This scenario generates negatively correlated resource profiles and represents ecological trade-offs or specialization constraints.

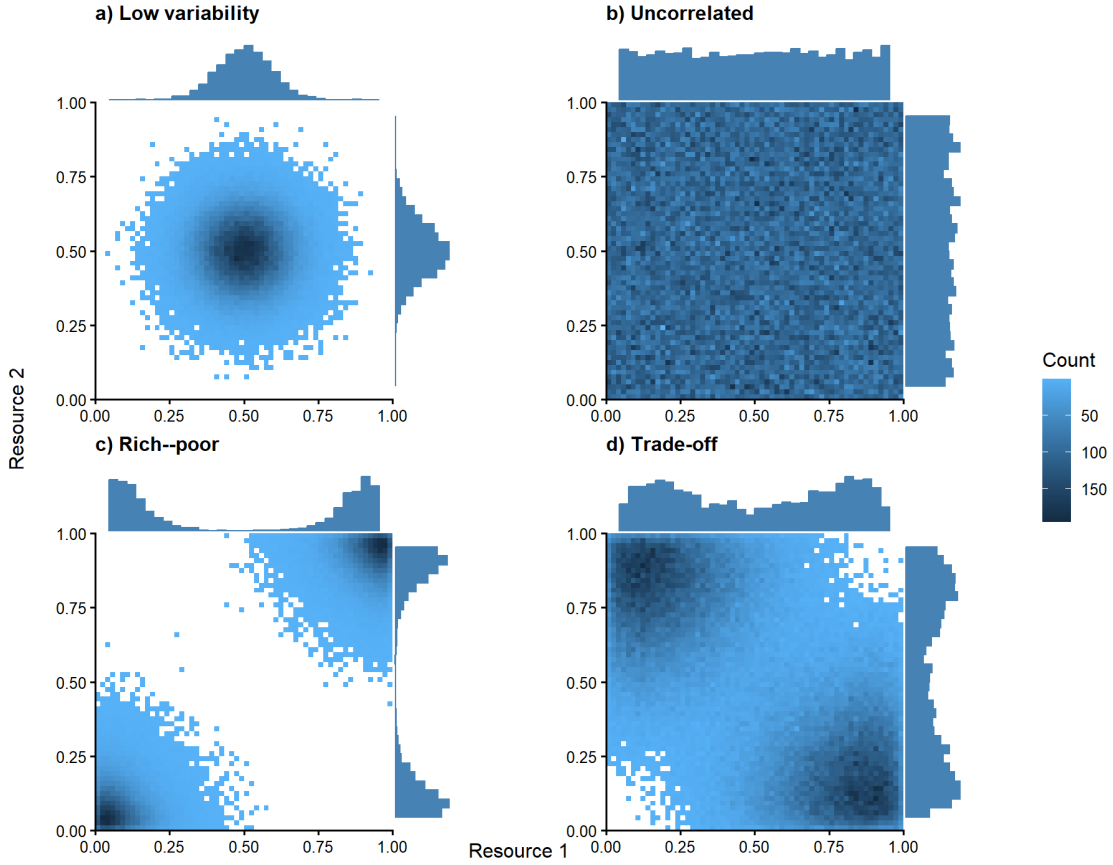


Figure 2: Environmental variability conditions. Joint distributions of initial resource endowments (r_1, r_2) across agents, shown as heatmaps (120,000 samples; 60×60 bins), with marginal univariate distributions displayed as histograms (3,000 points). (a) Low variability. (b) Uncorrelated variability. (c) Rich-poor. (d) Trade-off.

2.3.3 Initial conditions

At the start of each simulation, the population is monomorphic with respect to both traits: the dissatisfaction threshold is set to $\lambda = 0$, so that all agents initially avoid interaction, and the belligerence trait is set to $b = 0.5$, so that agents are initially unbiased between conflict and trade when engaging. Traits then evolve through mutation and selection. Mutations occur independently for each trait with probabilities $\mu_b = 0.01$ for the belligerence trait and $\mu_\lambda = 0.01$ for the dissatisfaction threshold. Mutational effects are drawn from normal

distributions with mean 0 and standard deviations $\sigma_b = 0.02$ and $\sigma_\lambda = 0.02$, respectively. Trait values are constrained to lie within the interval $[0, 1]$.

Parameter	Description	Value
G	Population size	10'000
μ_b	Mutation rate (belligerence)	0.01
μ_λ	Mutation rate (threshold)	0.01
σ_b	Mutation SD (belligerence)	0.02
σ_λ	Mutation SD (threshold)	0.02
w_i	Fecundity function	$\min(r_{1i}, r_{2i})$

Table 3: Summary of the model parameters used in the analysis.

2.3.4 Simulation comparisons

We compare the results obtained under the low-variability environment with those obtained under the three high-variability conditions (Uncorrelated, Trade-off, and Rich-poor), based on the trajectories of the belligerence trait b and the dissatisfaction threshold λ , as well as on the frequencies of avoidance, conflict, and trade.

2.4 Materials

All agent-based simulations were implemented in Julia (v1.12.2). For each environmental condition and model variant, simulations were run for 200,000 generations with a fixed population size of 10,000 agents. Each parameter combination was replicated 20 times.

For each simulation, time series of trait values and proportions of interactions were recorded across generations. Both between-run variability across replicates and within-population variability within generations were quantified.

Simulation outputs were exported as CSV files and processed in R (v4.4.2). Data management and visualisation were performed using the packages `data.table`, `dplyr`, `tidyr`, `ggplot2`, `scales`, and `ggnewscale`. All source code and analysis scripts are publicly available in the associated GitHub repository, together with a reproducible specification of package versions (<https://github.com/ayoubhartaoui/Agent-Based-Simulation>).

3 Results

We investigated the evolution of the dissatisfaction threshold λ and belligerence b under several resource-variability scenarios, and examined the resulting variation in action frequencies within the population. In particular, we analysed how the nature of resource variation affects the balance between conflict (war and raid) and cooperative exchange (trade). As a baseline, we first consider a low-variability scenario and then compare it to scenarios with higher variability, including uncorrelated, rich–poor, and trade-off resource distributions.

3.1 Low-variability environment

In the low-variability environment, the dissatisfaction threshold λ increases rapidly from its initial value of 0 and then stabilises at a low level (around ~ 0.15) across all conflict-cost scenarios (Figure 3c). This equilibrium value shows little variation across replicate simulations. Within populations, the distribution of λ remains narrow at both low and high conflict costs, with variances close to zero and no evidence of bimodality (Figure S4 and Figure S5 c)

Because resource endowments are tightly concentrated around 0.5, the low value of λ lead to a low frequency of engagement. On average, more than 99% of agents avoid interaction, and fewer than 1% engage across all cost scenarios (Figure S1 a).

Belligerence b fluctuates around intermediate values and does not exhibit clear convergence or separation between conflict-cost treatments (Figure 3a). Within-population variation in belligerence remains limited, the variance of b stays low through time (Figure S4 and Figure S5 a). The joint distribution of (b, λ) forms a single continuous cloud, with no distinct clusters across cost scenarios, showing no clear sign of stable trait polymorphism within populations (Figure S4 and Figure S5 d).

Action frequencies display variation across replicate simulations. Under low variability and low conflict costs, war and trade occur at comparable average frequencies, with a slight predominance of trade (Figure 4a). This variation in actions arises despite limited within-

population variation in trait values.

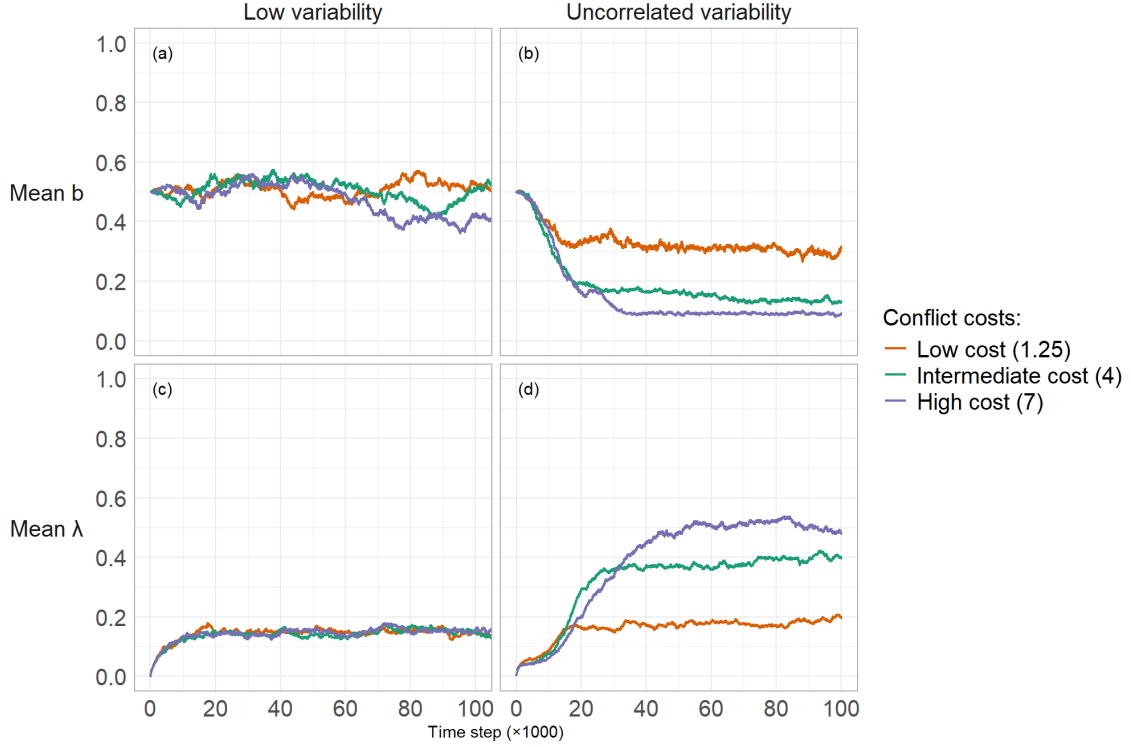


Figure 3: Time series of mean belligerence b (top row) and mean dissatisfaction threshold λ (bottom row) under low-variability (left column) and uncorrelated-variability (right column) environments. Lines show means across 20 replicate simulations for different conflict-cost scenarios. Simulations run for 100,000 generations. Panels show (a) b under low variability, (b) b under uncorrelated variability, (c) λ under low variability, and (d) λ under uncorrelated variability.

3.2 High-variability environments

3.2.1 Uncorrelated variability environment

Under uncorrelated variability, the dissatisfaction threshold λ increases from its initial value of 0 and reaches higher equilibrium values than under low variability, with final values depending on conflict costs (Figure 3d). At high conflict costs, λ exceeds 0.5, whereas at

low costs it remains lower but still above the levels observed under low variability. While interaction remains rare in the low-variability scenario, under uncorrelated variability a large fraction of the population engages, rising to more than half of the agents at high conflict costs (up to $\sim 56\%$ see Figure S1 b).

Within populations, the distribution of λ remains unimodal across conflict-cost scenarios, with no evidence for distinct clusters or stable polymorphism (Figure S6 c and Figure S7 c). Belligerence b declines with increasing conflict costs (Figure 3b). At low conflict costs, mean b remains relatively high (around $b \approx 0.3$), whereas at intermediate and high costs it converges toward lower values. Within-population variation in b is larger than under low variability, but histogram distributions remain unimodal across all cost scenarios (Figure S6 b and Figure S7 b). The joint distribution of (b, λ) forms a continuous cloud without distinct clusters (Figure S6 d and Figure S7 d).

These trait patterns are reflected in the distribution of actions among non-avoiding agents (Figure 4). Under uncorrelated variability, trade constitutes the dominant interaction across all conflict-cost scenarios, and its proportion increases with conflict costs. At low conflict costs, trade already accounts for approximately 50% of interactions among non-avoiders, while war falls below 12% (Figure 4a). This contrasts with the low-variability environment, where war and trade occur at comparable frequencies at the same cost level. At high conflict costs, trade rises to around 80% and war declines below 5% under uncorrelated variability, whereas under low variability median trade remains below 60% and median war below 10% (Figure 4c). Thus, increasing conflict costs reduce the frequency of war in both environments, but this reduction is substantially stronger when resource variability is high, even though trait distributions remain unimodal and do not indicate different behavioural strategies.

Importantly, the decline of war does not coincide with the disappearance of belligerent traits. Conflict-prone strategies persist at low but non-zero frequencies, as indicated by the continued presence of raid. As a result, war (conflict–conflict encounters) declines more rapidly than raid, which corresponds to interactions between conflict-prone and trade-prone agents

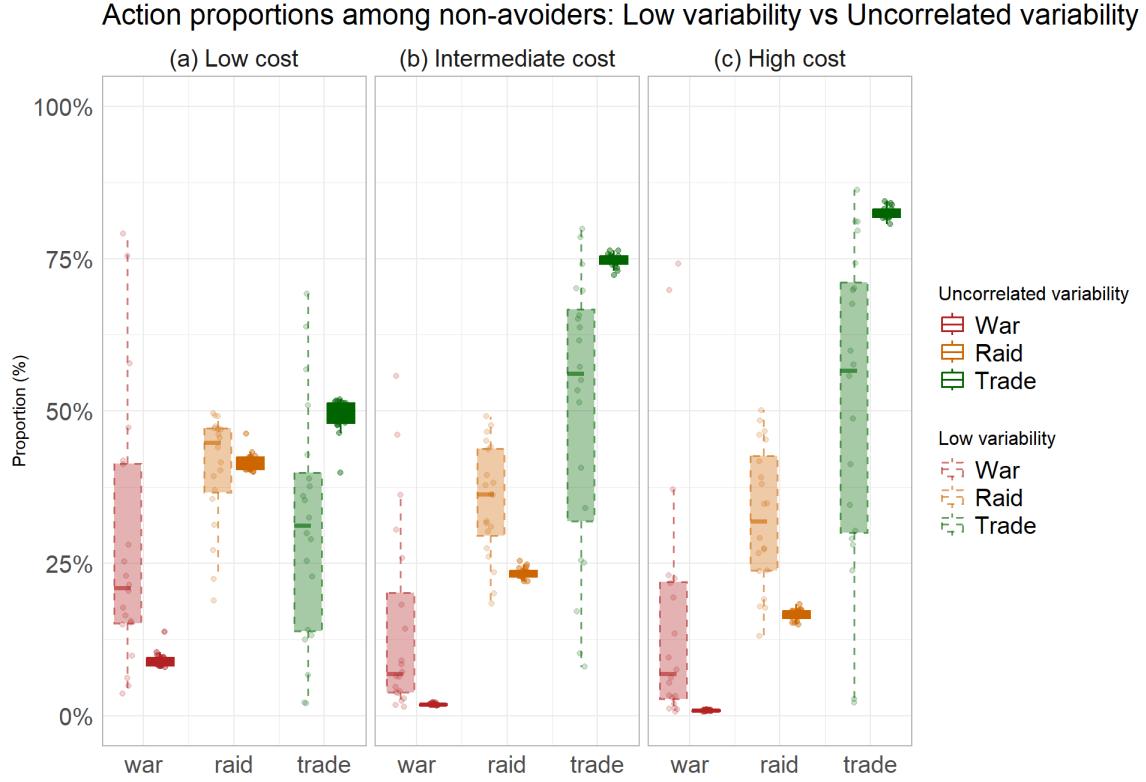


Figure 4: Action proportions among non-avoiding agents in low- and uncorrelated-variability environments under (a) low (1.25), (b) intermediate (4), and (c) high (7) conflict costs. Boxplots show the distribution of action frequencies across replicate simulations, with points indicating individual replicates. Proportions are calculated conditional on engagement, such that war, raid, and trade sum to 100%. Values represent time-averaged proportions over the final 10,000 generations of each simulation (generations 190,000–200,000).

3.2.2 Rich-poor variability environment

We next consider the rich–poor variability scenario, in which half of the population receives high values of both resources and the other half receives low values of both resources.

At low conflict costs, trade already accounts for approximately 75% of interactions among non-avoiders, while war occurs at very low frequencies. At intermediate and high conflict costs, action frequencies stabilise around a similar pattern, with roughly 85–90%

trade, around 10–15% raid, and almost no war on average (Figure 5).

These patterns are reflected in the evolution of traits (Figure 6). Mean belligerence remains low across all conflict-cost scenarios, stabilising around values close to 0.1 and showing little variation with cost. The dissatisfaction threshold λ also stabilises at low to intermediate values, around 0.2 at low conflict costs and around 0.3 at intermediate and high costs only agents with resources endowments lower than this act. Engagement therefore remains restricted to agents experiencing resource deprivation. Across conflict-cost treatments, approximately 18–20% of agents engage in interaction (Figure S1 c), higher proportion than under low variability, but lower than the highest engagement levels observed under uncorrelated variability.

Within populations, trait distributions remain unimodal across conflict-cost scenarios. Histograms of both b and λ show no evidence of bimodality, and the joint distribution of (b, λ) forms a single continuous cloud, indicating the absence of stable trait polymorphism (Figure S8 and Figure S9).

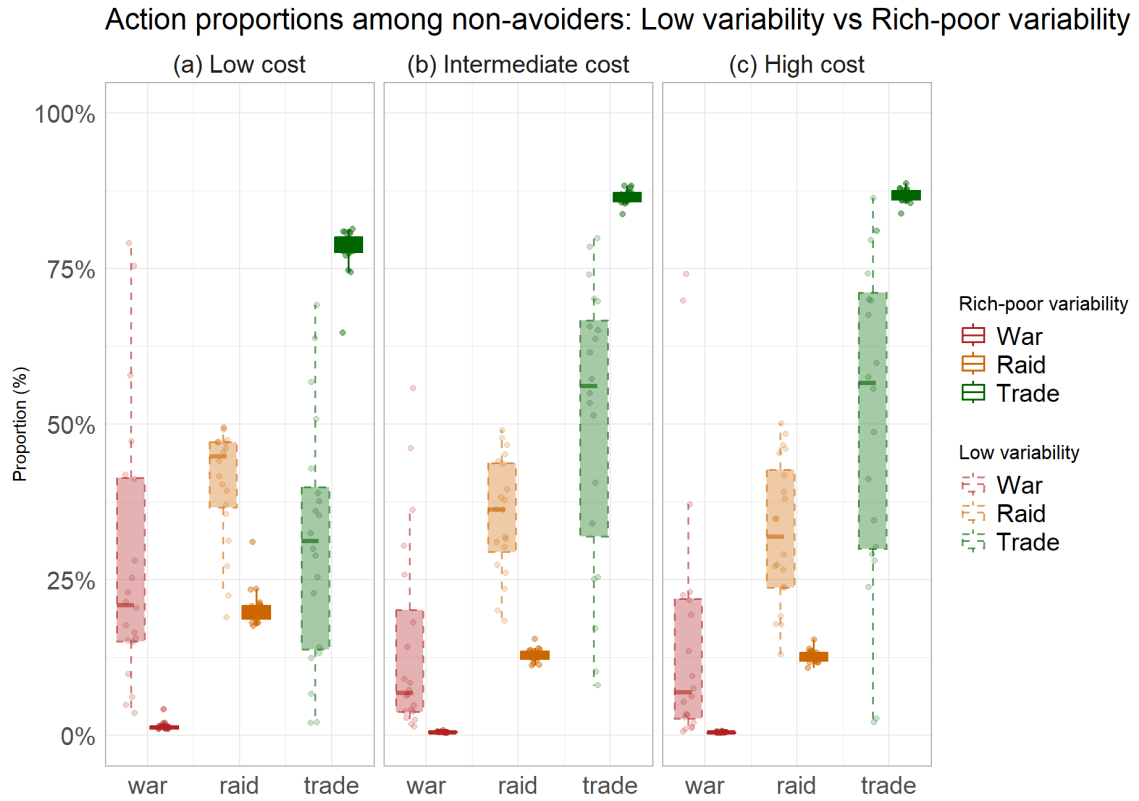


Figure 5: Action proportions among non-avoiding agents in low-variability and rich-poor environments under (a) low (1.25), (b) intermediate (4), and (c) high (7) conflict costs. Box-plots show the distribution of action frequencies across replicate simulations, with points indicating individual replicates. Proportions are calculated conditional on engagement, such that war, raid, and trade sum to 100%. Values represent time-averaged proportions over the final 10,000 generations of each simulation (generations 190,000–200,000).

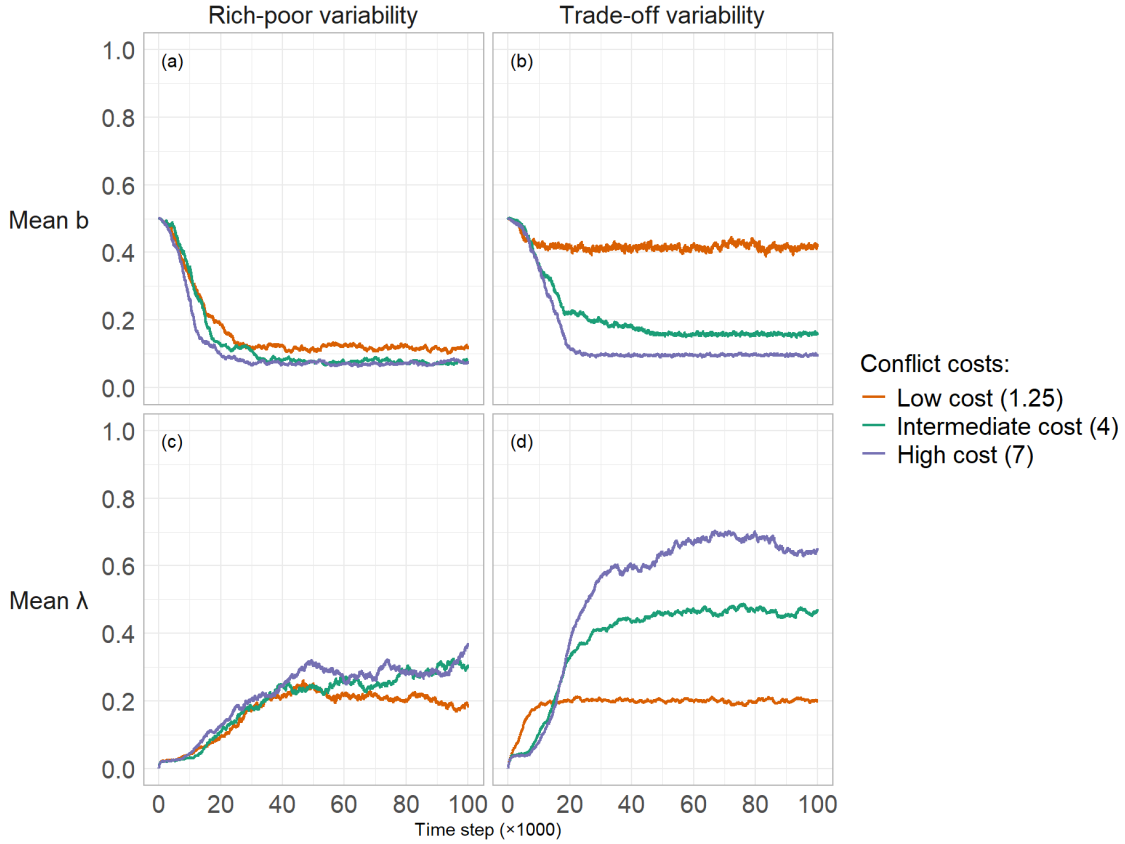


Figure 6: Time series of mean belligerence b (top row) and mean dissatisfaction threshold λ (bottom row) under rich–poor variability (left column) and trade-off variability (right column). Lines show means across 20 replicate simulations for different conflict-cost scenarios. Simulations were run for 100,000 generations. Panels show (a) b under low variability, (b) b under uncorrelated variability, (c) λ under low variability, and (d) λ under uncorrelated variability

3.2.3 Trade-off variability environment

Trade-off variability is associated with the highest levels of engagement among all variability scenarios. The dissatisfaction threshold λ increases with conflict costs and reaches high equilibrium values (Figure 6d). The fraction of agents that engage in interaction rises sharply with costs, from approximately 24% at low conflict costs to 77% at intermediate

costs, and up to 96% at high conflict costs (Figure S1 d).

At low conflict costs, the distribution of actions among non-avoiding agents differs from other high-variability environments (Figure 7). Raid constitutes the most frequent action, accounting for nearly 50% of interactions, while trade remains common but slightly less frequent (around 35%), and war occurs at lower frequencies (around 15%). This contrasts with uncorrelated and rich–poor variability, where trade already reached high frequencies at low conflict costs.

As conflict costs increase, this distinction disappears. At intermediate and high conflict costs, trade–off variability converges toward the same interaction pattern observed under other high-variability scenarios. Trade becomes the highest frequency interaction among non-avoiding agents, war collapses to near-zero frequencies, and raid persists only at low but stable levels.

Trait dynamics under trade–off variability closely mirror those observed under uncorrelated variability. Mean belligerence b declines with increasing conflict costs (Figure 6b), while the dissatisfaction threshold λ increases and stabilises at high values (Figure 6d). Within populations, trait distributions remain unimodal across all conflict-cost scenarios, with no evidence of stable polymorphism in either b or λ (Figure S10 and Figure S11).

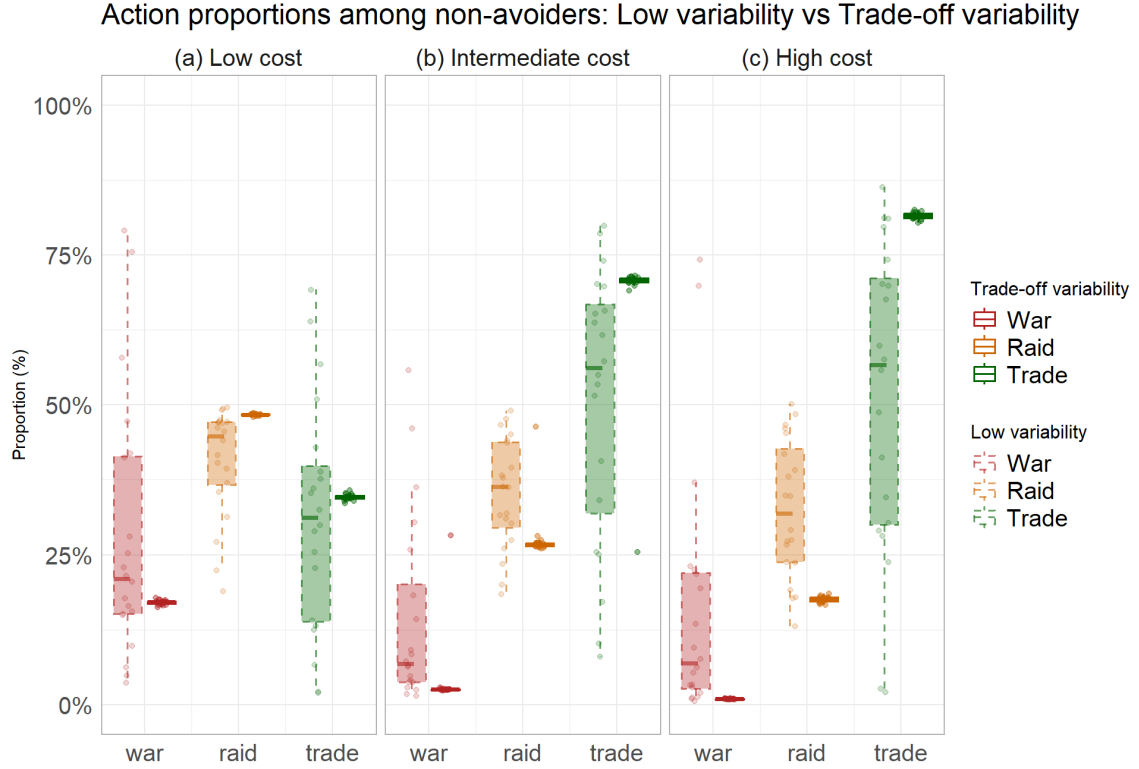


Figure 7: Action proportions among non-avoiding agents in the low-variability and trade-off environments under (a) low (1.25), (b) intermediate (4), and (c) high (7) conflict costs. For each cost scenario, boxplots show the distribution across replicate simulations and points indicate replicate values. Proportions are calculated conditional on engagement, such that raid, war, and trade jointly sum to 100%. Values correspond to time-averaged proportions computed over the final 10,000 generations of each simulation (generations 190,000–200,000).

3.2.4 Avoiders cannot escape conflict

As described in Section 2.2.1, we also consider a modified interaction rule in which avoidance no longer guarantees escape from conflict. Under this alternative rule, agents that choose avoidance are still exposed to conflictual encounters when paired with a belligerent opponent and must relinquish their resource endowments. This modification allows us to assess how constraining avoidance alters the behavioural and evolutionary outcomes

described above.

Constraining avoidance alters trait dynamics relative to the baseline model. Across all environments and conflict-cost scenarios, the dissatisfaction threshold λ evolves to high values, with the lowest equilibrium values remaining above 0.7 even under low variability and high conflict costs (Figure S12). Also, interactions are no longer restricted with engagement exceeding 90% in most cases (Figure S2).

This change is accompanied by a separation between low- and high-cost regimes in belligerence. At low conflict costs, mean belligerence reaches substantially higher values than in the baseline model, rising to approximately 0.6–0.8 depending on the environment, whereas it remains low at intermediate and high costs. Such high levels of belligerence were not observed previously, where belligerence never exceeded 0.4.

The effect of resource variability on belligerence is preserved for uncorrelated and trade-off variability (Figure S12 and Figure S13). In these environments, increasing variability continues to reduce belligerence by comparable magnitudes (approximately 0.1–0.2). This mirrors the patterns observed in the baseline model where those two environments lowered belligerence by 0.1.

The rich–poor environment, however, behaves differently under constrained avoidance. At low conflict costs, belligerence reaches very high values (around 0.8), and the dissatisfaction threshold λ converges to values close to one, implying that nearly all agents engage in interactions (Figure S13). In this regime, war becomes the predominant interaction, accounting for approximately 55–60% of encounters (Figure S15). As conflict costs increase, both belligerence and λ collapse sharply, as well as war frequency. Thus, under this interaction rule, social inequality increase conflict frequency when conflict is inexpensive but reduces it once costs rise.

Precisely, across uncorrelated and trade–off environments, higher variability is associated with reduced conflictual actions and increased frequencies of cooperative exchange across all conflict-cost scenarios (Figure S14 and Figure S16). Although constraining avoidance leads to higher mean belligerence and higher levels of conflict compared to the baseline model overall, the effect of high resource variability remains unchanged, higher

variability is still associated with fewer conflictual interactions and more cooperative exchange. The only exception occurs in the rich–poor environment at low conflict costs, where war dominates (Figure S15). We considered this variant to address a question raised by the baseline rich–poor scenario, where interactions are largely restricted to poor agents. To answer if well-endowed agents cannot avoid interaction, does high variability increase conflict by exposing them to exploitation.

3.2.5 Complementary resources

Finally, we examine how altering the biological interpretation of resource complementarity affects our results. In the main model, reproductive success is determined by the scarcest resource, reflecting an organismic view in which fitness is constrained by essential limitations. To assess the robustness of our conclusions, we consider an alternative formulation in which fecundity follows a Cobb–Douglas utility function, capturing situations in which multiple resources jointly contribute to reproduction and are imperfect substitutes.

At the level of action frequencies, uncorrelated and trade-off variability under the Cobb–Douglas formulation behave similarly to trade-off variability in the baseline model (Figure S19 and Figure S21). This contrasts with uncorrelated variability in the baseline model, where trade already exceeds raid at low costs. As conflict costs increase, however, this distinction disappears. At intermediate and high costs, uncorrelated and trade-off variability converge toward the same interaction frequencies trend observed in other high-variability environments, with trade representing the highest played interaction among non-avoiding agents, war collapsing to near-zero frequencies, and raid persisting at low but stable levels. Rich–poor variability continues to promote cooperative exchange across all cost scenarios, as in the baseline model (Figure S20).

Although mean action frequencies follow the same trends as in the limiting–resource model, we see that the Cobb–Douglas formulation show greater dispersion across replicate simulations. Under high variability, engagement rises only to low levels (10–25%, see Figure S3), whereas in the baseline model it reaches higher engagement level. Action proportions show greater dispersion across replicate simulations than under the limit-

ing–resource formulation.

4 Discussion

The central question of this study is how environmental stochasticity shapes the evolution of conflict and cooperation when both strategies are available responses to competition over finite resources. We addressed this question by examining how different forms of resource variability influence engagement, behavioural traits, and the resulting balance between conflictual and cooperative actions. Across scenarios, increasing resource variability led to higher levels of engagement and an increase in cooperative exchange.

In low-variability environments, most agents avoided interaction (Figure S1a), and among the few agents that did engage, war and trade occurred at comparable frequencies, particularly at low conflict costs. By contrast, under high variability, interactions became more frequent and were increasingly dominated by cooperative exchange, while war collapsed to near-zero frequencies and conflict persisted only at low levels, even when conflict costs were minimal (Figure 4, Figure 7 and Figure 5).

Our finding that increased resource variability leads to less conflictual interactions may seem at odds with the common expectation that harsh or unpredictable environments favour aggression (Abbink et al., 2018; De Courson et al., 2023). In our model, this pattern arises from the evolution of selective engagement. Engagement decisions are mediated by the dissatisfaction threshold λ , which determines whether agents interact or avoid interaction based on their current resource state.

Under high variability, interaction becomes increasingly conditional, agents engage only when their resources fall below a critical level (Figure 6). This filtering strongly shapes the composition of interacting agents. When avoidance is constrained, engagement becomes widespread across all environments, but the effect of high resource variability on interaction frequency persists (Figure S2 and Figure S14). This shows that the influence of environmental variability is not driven by differences in the proportion of agents that engage, but by differences in who engages. Under low variability, interacting agents tend

to have similar resource endowments, whereas under high variability interactions involve agents that differ strongly in resource levels. By reshaping the composition of interacting pairs, high resource variability continues to tip the balance from conflict to higher level of cooperative exchange than observed under low variability.

When selective engagement is present, interactions concentrate among poorly endowed agents. Across high-variability environments, agents that engage tend to hold low levels of both resources, while better-endowed agents predominantly avoid interaction. In this context, conflict is associated with limited gains and high evolutionary risks. Seizing resources from another poorly endowed agent yields small absolute benefits, while losing a conflict prevents reproduction. As a result, cooperative exchange dominates among interacting agents, while conflict is strategically disfavoured.

This disadvantage of conflict is frequency-dependent. As belligerent strategies become more common, conflict–conflict encounters increase and are strongly penalised. Consequently, war collapses to near-zero frequencies, while raid persists at low but stable levels (Figure 5). Resource variability also increases engagement by raising the frequency with which agents experience low resource states. Under high variability, a substantially larger fraction of the population engages in interaction than under low variability, even when conflict costs are low (Figure S1).

This pattern is consistent with empirical work on state-dependent risk responses, which shows that resource scarcity can increase individuals’ willingness to engage in risky actions rather than remain inactive. In particular, Courson et al. document a non-monotonic relationship between perceived resources and risk-taking, with engagement increasing as resources decline toward critical thresholds (De Courson et al., 2025). In our model, environmental variability similarly promotes engagement under scarcity, while selection favours cooperative exchange as the least risky way to remain active.

Importantly, these patterns arise without the emergence of stable polymorphisms in behavioural traits. Across environments and conflict-cost regimes, within-population distributions of both belligerence and the dissatisfaction threshold remain unimodal (Figures S6–S11). This indicates that variation in observed interactions primarily reflects

state-dependent action driven by the resource endowments and engagement decision choice, rather than the coexistence of distinct behavioural types.

Rich–poor variability illustrates particularly clearly how shifts in who interacts alter the strategic balance between conflict and cooperation. In this environment, interactions occur almost exclusively among resource-poor agents as the dissatisfaction threshold is low (Figure 6a,c). When two poorly endowed agents interact, the total amount of resources at stake is small (e.g. on the order of 0.4 when both hold resources around 0.2), while conflict costs remain unchanged (e.g. 1.25 at low cost). Relative to environments in which interactions involve larger resource endowments (e.g. total resources on the order of 1.4 when both agents hold around 0.7), the situation where only poor interact is equivalent to facing higher effective conflict costs. Rich–poor variability therefore effectively rescales conflict costs relative to gains, making conflict unattractive even when costs are considered as “low”.

Trade–off variability highlights the limits of this mechanism at low conflict costs. In this environment, gain differences between interaction types remain relatively symmetric when conflict is less costly, and raid dominates interactions at low costs (Figure 7). As conflict costs rise, however, trade–off variability converges toward the same cooperative regime observed under other high-variability environments. Non-belligerent actions become increasingly favoured, war collapses, and trade dominates interactions among non-avoiding agents.

The trend identified here is consistent with archaeological evidence linking increased environmental variability to changes in patterns of interaction and exchange. Periods of heightened ecological variability in East Africa around 300,000 BP coincide with increased raw material transfer distances, suggesting expanded mobility and exchange networks rather than increased isolation or conflict (Potts, 2013). Our results provide a possible interpretation of this association, when fitness is constrained by minimal resource requirements, high variability increase engagement incentives and favours cooperative exchange.

This trend also align with recent theoretical and experimental work showing that re-

source variability can promote cooperation. In particular, Inaba and Akiyama (2024) show that variability alone can sustain cooperative outcomes, while experimental studies of common-pool resource dilemmas report increased engagement and reciprocal resource sharing under environmental shocks (Safarzynska, 2018). By explicitly allowing for avoidance, exploitation, and violent conflict, our model extends these approaches and shows how cooperative exchange can reach high frequencies even when conflict remains a viable strategic option.

However effects of resource variability depend critically on how conflict costs are distributed. When avoidance is constrained and conflict is inexpensive, the same asymmetries that promote cooperation in the baseline model can instead amplify conflict in the rich–poor environment (Figure S15). Poor but highly belligerent agents gain access to well-endowed opponents who cannot withdraw, increasing the returns to resource seizing and leading to elevated levels of conflict. This relationship mirrors experimental and theoretical findings showing that behaviour changes sharply around critical resource thresholds. Models and experiments in which individuals seek to remain above a desperation threshold find that moderate inequality can increase conflict, whereas extreme inequality may reduce it through resignation or disengagement (De Courson et al., 2023; Abbink et al., 2018).

We also examined whether our conclusions depend on how resources translate into reproductive success. In the main model, fecundity follows a limiting-resource rule, consistent with an organismic view in which growth is constrained by the scarcest essential resource. We contrasted this with a Cobb–Douglas specification, commonly used in economic models, in which fecundity depends on the complementary accumulation of both resources (Cobb and Douglas, 1928). Across environments, action frequencies follow the same broad trends observed under the limiting-resource model, including the shift toward cooperative exchange under high variability and increasing conflict costs (Figure S19 – Figure S21). However, the Cobb–Douglas formulation is associated with greater between-run stochasticity and higher avoidance proportions. This difference likely reflects weakens incentive to act relative to the minimum rule.

These findings speak to a broader and still unresolved question in human evolutionary

history: the coexistence of cooperation and conflict between unrelated individuals, often referred to as the “paradox of peace and war” (Fry, 2013, 2025; Glowacki, 2024; Wilson, 2025). Rather than assuming that humans are inherently peaceful or violent, our results may help clarify how different interaction patterns can be favoured.

Our model reproduces this pattern under the lens of resource inequality. When there is no strong incentive to engage, such as severe resource scarcity that constrains fecundity, agents predominantly avoid interaction, even though engaging in conflict could, in principle, allow them to seize additional resources and increase reproductive success. Under these conditions, selection does not favour increased risk-taking or high engagement thresholds. When interactions do occur, they are concentrated among poorly endowed agents, for whom cooperative exchange is strategically favoured over conflict. In this sense, high environmental variability combined with costly conflict can generate interaction patterns broadly consistent with scenarios emphasising avoidance and limited warfare as advocated by Fry (2013).

At the same time, our results do not imply that conflict was absent or evolutionarily unimportant. If interactions are unavoidable, conflict costs are low, or resource access is strongly asymmetric, belligerent strategies can spread and conflict becomes prevalent. This leaves open the possibility that warfare in past societies may have been more episodic, context-dependent, or less costly than is often assumed, as suggested by recent work (Xiao et al., 2026).

To conclude, it is important to emphasize that the present model deliberately relies on simplified assumptions. We consider a well-mixed population without relatedness, repeated interactions, or complex group formation, and conflict outcomes are determined by a stochastic winner-takes-all rule rather than by more elaborate contest success functions. These choices were made intentionally. The aim of this model is not to provide a fully realistic account of social conflict or cooperation, but rather to isolate a trend linking environmental variability, selective engagement, and the balance between conflict and cooperative exchange.

By stripping away additional layers of complexity, the model allows us to identify drivers

of these outcomes while remaining transparent and interpretable. In this sense, the present framework should be viewed as a baseline or stepping stone rather than a comprehensive theory. Incorporating spatial structure, repeated interactions, complex group formation, or more detailed conflict resolution mechanisms represents promising direction for future work aimed at better understanding the evolutionary dynamics of peace and war.

5 Acknowledgements

I am deeply grateful to my supervisor, Miguel Dos Santos, for his guidance and support throughout this project, from the early stages of experimental design to code optimisation and comments on previous versions of the manuscript. I also thank my director, Charles Mullon, for his invaluable advice, comments on previous versions of the manuscript, contributions to model design, and constant encouragement during the development of this work. I am thankful to Cédric Perret for his availability and assistance during the development of the model. I also thank the members of the Laboratory of Theoretical Evolutionary Ecology for their constructive feedback and helpful discussions. Finally, I thank my friend Axel Fleury, assistant doctoral researcher at the Institute of Finance at the University of Neuchâtel, for informal discussions and support.

References

- Abbink, K., Masclet, D., and Mirza, D. (2018). Inequality and inter-group conflicts: Experimental evidence. *Social Choice and Welfare*, 50(3):387–423.
- Almstead, D. K., Savagian, A. G., Smith, M. G., and Riehl, C. (2021). Inter-group conflict in a cooperatively breeding bird: New insights into “home field advantage”. *Ethology*, 127(4):372–378.
- Bausch, A. W. (2014). Evolving intergroup cooperation. *Computational and Mathematical*

- Organization Theory*, 20(4):369–393. Publisher: Springer Science and Business Media LLC.
- Bowles, S. (2009). Did Warfare Among Ancestral Hunter-Gatherers Affect the Evolution of Human Social Behaviors? *Science*, 324(5932):1293–1298. Publisher: American Association for the Advancement of Science (AAAS).
- Böhm, R., Rusch, H., and Baron, J. (2020). The psychology of intergroup conflict: A review of theories and measures. *Journal of Economic Behavior & Organization*, 178:947–962. Publisher: Elsevier BV.
- Choi, J.-K. and Bowles, S. (2007). The Coevolution of Parochial Altruism and War. *Science*, 318(5850):636–640. Publisher: American Association for the Advancement of Science (AAAS).
- Cobb, C. W. and Douglas, P. H. (1928). A theory of production. *The American Economic Review*, 18(1):139–165. Supplement, Papers and Proceedings of the Fortieth Annual Meeting of the American Economic Association.
- De Courson, B., Frankenhuis, W. E., and Nettle, D. (2025). Poverty is associated with both risk avoidance and risk taking: empirical evidence for the desperation threshold model from the UK and France. *Proceedings of the Royal Society B: Biological Sciences*, 292(2040):20242071.
- De Courson, B., Frankenhuis, W. E., Nettle, D., and Van Gelder, J.-L. (2023). Why is violence high and persistent in deprived communities? A formal model. *Proceedings of the Royal Society B: Biological Sciences*, 290(1993):20222095.
- De Dreu, C. K., Gross, J., Fariña, A., and Ma, Y. (2020). Group Cooperation, Carrying-Capacity Stress, and Intergroup Conflict. *Trends in Cognitive Sciences*, 24(9):760–776. Publisher: Elsevier BV.

- Dyble, M. (2021). The evolution of altruism through war is highly sensitive to population structure and to civilian and fighter mortality. *Proceedings of the National Academy of Sciences*, 118(11). Publisher: Proceedings of the National Academy of Sciences.
- Frank, S. A. (2007). All of life is social. *Current Biology*, 17(16):R648–R650.
- Fry, D. P., editor (2013). *War, peace, and human nature: the convergence of evolutionary and cultural views*. Oxford University Press, New York.
- Fry, D. P. (2025). The quest for scientific objectivity: comment on Luke Glowacki’s (2024) “The controversial origins of war and peace: Apes, foragers, and human evolution”. *Evolution and Human Behavior*, 46(1):106650. Publisher: Elsevier BV.
- Glowacki, L. (2024). The evolution of peace. *Behavioral and Brain Sciences*, 47. Publisher: Cambridge University Press (CUP).
- Gross, J., Méder, Z. Z., De Dreu, C. K., Romano, A., Molenmaker, W. E., and Hoenig, L. C. (2023). The evolution of universal cooperation. *Science Advances*, 9(7). Publisher: American Association for the Advancement of Science (AAAS).
- Hölldobler, B. and Wilson, E. O. (2009). *The superorganism: the beauty, elegance, and strangeness of insect societies*. W.W. Norton, New York, 1st ed edition.
- Inaba, M. and Akiyama, E. (2024). Environmental variability promotes the evolution of cooperation among geographically dispersed groups on dynamic networks. arXiv:2411.12348 [physics].
- Kalai, E. and Smorodinsky, M. (1975). Other solutions to nash’s bargaining problem. *Econometrica*, 43(3):513–518.
- Krebs, J. R., Davies, N. B., and Parr, J. (1982). *An introduction to behavioural ecology*. Blackwell, Oxford, reprinted edition.
- Lehmann, L. and Feldman, M. W. (2008). War and the evolution of belligerence and bravery. *Proceedings of the Royal Society B: Biological Sciences*, 275(1653):2877–2885.

- Li, Y.-H. and Tian, X. (2012). Quorum Sensing and Bacterial Social Interactions in Biofilms. *Sensors*, 12(3):2519–2538. Publisher: MDPI AG.
- Liebig, J. v. (1840). *Die organische Chemie in ihrer Anwendung auf Agricultur und Physiologie*. Vieweg, Braunschweig, 1 edition. URN: urn:nbn:de:kobv:b4-200905193607. Staatsbibliothek zu Berlin – Preußischer Kulturbesitz, Signatur SBB-PK, Ov 15534a R.
- Metz, M. C., Vucetich, J. A., Smith, D. W., Stahler, D. R., and Peterson, R. O. (2011). Effect of Sociality and Season on Gray Wolf (*Canis lupus*) Foraging Behavior: Implications for Estimating Summer Kill Rate. *PLoS ONE*, 6(3):e17332.
- Micheletti, A. J. C., Ruxton, G. D., and Gardner, A. (2018). Why war is a man’s game. *Proceedings of the Royal Society B: Biological Sciences*, 285(1884):20180975. Publisher: The Royal Society.
- Potts, R. (2013). Hominin evolution in settings of strong environmental variability. *Quaternary Science Reviews*, 73:1–13.
- Rodrigues, A. M. M., Barker, J. L., and Robinson, E. J. H. (2023). The evolution of intergroup cooperation. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 378(1874). Publisher: The Royal Society.
- Rusch, H. (2014). The Two Sides of Warfare: An Extended Model of Altruistic Behavior in Ancestral Human Intergroup Conflict. *Human Nature*, 25(3):359–377. Publisher: Springer Science and Business Media LLC.
- Rusch, H. (2022). Modelling behaviour in intergroup conflicts: a review of microeconomic approaches. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 377(1851). Publisher: The Royal Society.
- Safarzynska, K. (2017). Intergroup cooperation prevents resource exhaustion but undermines intra-group cooperation in the common-pool resource experiment. *Ecology and Society*, 22(4). Publisher: Resilience Alliance, Inc.

- Safarzynska, K. (2018). The Impact of Resource Uncertainty and Intergroup Conflict on Harvesting in the Common-Pool Resource Experiment. *Environmental and Resource Economics*, 71(4):1001–1025. Publisher: Springer Science and Business Media LLC.
- Sanjuán, R. (2021). The Social Life of Viruses. *Annual Review of Virology*, 8(1):183–199. Publisher: Annual Reviews.
- Sankey, D. W. E., Hunt, K. L., Croft, D. P., Franks, D. W., Green, P. A., Thompson, F. J., Johnstone, R. A., and Cant, M. A. (2022). Leaders of war: modelling the evolution of conflict among heterogeneous groups. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 377(1851). Publisher: The Royal Society.
- Smirnov, O., Arrow, H., Kennett, D., and Orbell, J. (2007). Ancestral War and the Evolutionary Origins of “Heroism”. *The Journal of Politics*, 69(4):927–940. Publisher: University of Chicago Press.
- Spikins, P., French, J. C., John-Wood, S., and Dytham, C. (2021). Theoretical and Methodological Approaches to Ecological Changes, Social Behaviour and Human Intergroup Tolerance 300,000 to 30,000 BP. *Journal of Archaeological Method and Theory*, 28(1):53–75. Publisher: Springer Science and Business Media LLC.
- Surbeck, M. and Hohmann, G. (2015). Social preferences influence the short-term exchange of social grooming among male bonobos. *Animal Cognition*, 18(2):573–579. Publisher: Springer Science and Business Media LLC.
- West, S. A., Griffin, A. S., and Gardner, A. (2007). Evolutionary Explanations for Cooperation. *Current Biology*, 17(16):R661–R672. Publisher: Elsevier BV.
- Wilson, M. L. (2025). Evidence and conceptual models for the evolution of war: a response to and. *Evolution and Human Behavior*, 46(3):106685. Publisher: Elsevier BV.
- Wrangham, R. W. and Glowacki, L. (2012). Intergroup Aggression in Chimpanzees and War in Nomadic Hunter-Gatherers: Evaluating the Chimpanzee Model. *Human Nature*, 23(1):5–29. Publisher: Springer Science and Business Media LLC.

Xiao, Q., Columbus, S., and Böhm, R. (2026). Self-serving intergroup aggression escalates and prevails over parochial cooperation. *Evolution and Human Behavior*, 47(1):106801.

S1 Supplementary Material

For clarity and conciseness, additional analyses and figures supporting the main results are presented here.

S1.1 Trade formulation

Consider two agents j and k with resource endowments (r_{1j}, r_{2j}) and (r_{1k}, r_{2k}) .

1) Limiting-resource (minimum-law) trade. In the limiting-resource formulation, fitness depends on the scarcest resource, so trade is implemented as a local rule that is only accepted if it increases each agent’s limiting resource. Define for each agent

$$m_j = \min(r_{1j}, r_{2j}), \quad m_k = \min(r_{1k}, r_{2k}), \quad (1)$$

and a “surplus” (resource imbalance)

$$s_j = \max(r_{1j}, r_{2j}) - \min(r_{1j}, r_{2j}), \quad s_k = \max(r_{1k}, r_{2k}) - \min(r_{1k}, r_{2k}). \quad (2)$$

Trade occurs only if agents are complementary, i.e. one is relatively richer in resource 1 while the other is relatively richer in resource 2:

$$(r_{2j} > r_{1j} \text{ and } r_{1k} > r_{2k}) \quad \text{or} \quad (r_{1j} > r_{2j} \text{ and } r_{2k} > r_{1k}). \quad (3)$$

When complementarity holds, each agent gives a fraction of its surplus to the other. The fraction is proportional to the partner’s limiting resource, so that the more constrained

agent receives relatively more. For instance, in the case $r_{2j} > r_{1j}$ and $r_{1k} > r_{2k}$,

$$t_j = \frac{m_k}{m_j + m_k}, \quad t_k = 1 - t_j, \quad (4)$$

and the amounts exchanged are

$$g_j = s_j t_j, \quad g_k = s_k t_k, \quad (5)$$

where g_j (resp. g_k) is the quantity of the abundant resource given by agent j (resp. k). Let (r'_{1j}, r'_{2j}) and (r'_{1k}, r'_{2k}) denote the post-trade endowments obtained by transferring these quantities between agents along their abundant resources. The trade is accepted if it strictly improves both limiting resources:

$$\min(r'_{1j}, r'_{2j}) > \min(r_{1j}, r_{2j}) \quad \text{and} \quad \min(r'_{1k}, r'_{2k}) > \min(r_{1k}, r_{2k}). \quad (6)$$

If either condition fails, no exchange occurs and endowments remain unchanged.

Cobb–Douglas trade formulation. The utility (fecundity proxy) of player i is defined as a Cobb–Douglas function of the two resources,

$$u_i(r_{1i}, r_{2i}) = r_{1i}^{1/2} r_{2i}^{1/2}. \quad (7)$$

Trade is restricted to exchanges that lie on the Pareto frontier of all feasible trades. In addition, we impose that the players' relative gains in utility are equal. This criterion is close in spirit to the Kalai–Smorodinsky bargaining solution, in which each player achieves the same proportion of their maximum feasible utility gain relative to the disagreement point. Here, we select an allocation that is Pareto-efficient and equalizes the players' relative improvements in utility.

Let the post-trade utilities of players A and B be denoted by u'_A and u'_B , respectively,

with

$$u'_A = u_A(r_{1A} + x, r_{2A} + y), \quad (8)$$

$$u'_B = u_B(r_{1B} - x, r_{2B} - y), \quad (9)$$

where x and y denote the net quantities of resources 1 and 2 transferred to player A. Feasibility requires

$$-r_{1,A} < x < r_{1B}, \quad -r_{2A} < y < r_{2B}. \quad (10)$$

The trade is chosen such that the relative increase in utility is the same for both players:

$$\frac{u_A(r_{1A} + x, r_{2A} + y)}{u_A(r_{1A}, r_{2A})} = \frac{u_B(r_{1B} - x, r_{2B} - y)}{u_B(r_{1B}, r_{2B})}. \quad (11)$$

Introducing the notation

$$\alpha = \frac{r_{2B}}{r_{2A}}, \quad \beta = \frac{r_{1A}}{r_{1B}}, \quad (12)$$

and solving the equal-relative-gain condition for x , assuming resources lie in the interval $(0, 1)$, yields

$$x^* = \frac{y \beta (1 + \alpha) r_{1B}}{y \beta - \alpha (r_{2A} + \beta r_{2A} + y)}. \quad (13)$$

Substituting x^* into the relative increase in utility of a player (e.g. player A, since both relative increases are equal) yields

$$u_A^* = \frac{u_A(r_{1A} + x^*, r_{2A} + y)}{u_A(r_{1A}, r_{2A})}. \quad (14)$$

We then select the value of y that maximizes u_A^* , subject to the constraint $u'_A \geq u_A$. This is obtained by solving

$$\frac{du_A^*}{dy} = 0, \quad (15)$$

which yields

$$y^* = r_{2A} \frac{\alpha \sqrt{\beta} - \sqrt{\alpha}}{\sqrt{\alpha} + \sqrt{\beta}}. \quad (16)$$

Finally, substituting y^* into the expression for x^* gives

$$x^* = r_{1,B} \frac{\sqrt{\beta} - \sqrt{\alpha} \beta}{\sqrt{\alpha} + \sqrt{\beta}}. \quad (17)$$

S1.2 Avoidance proportions

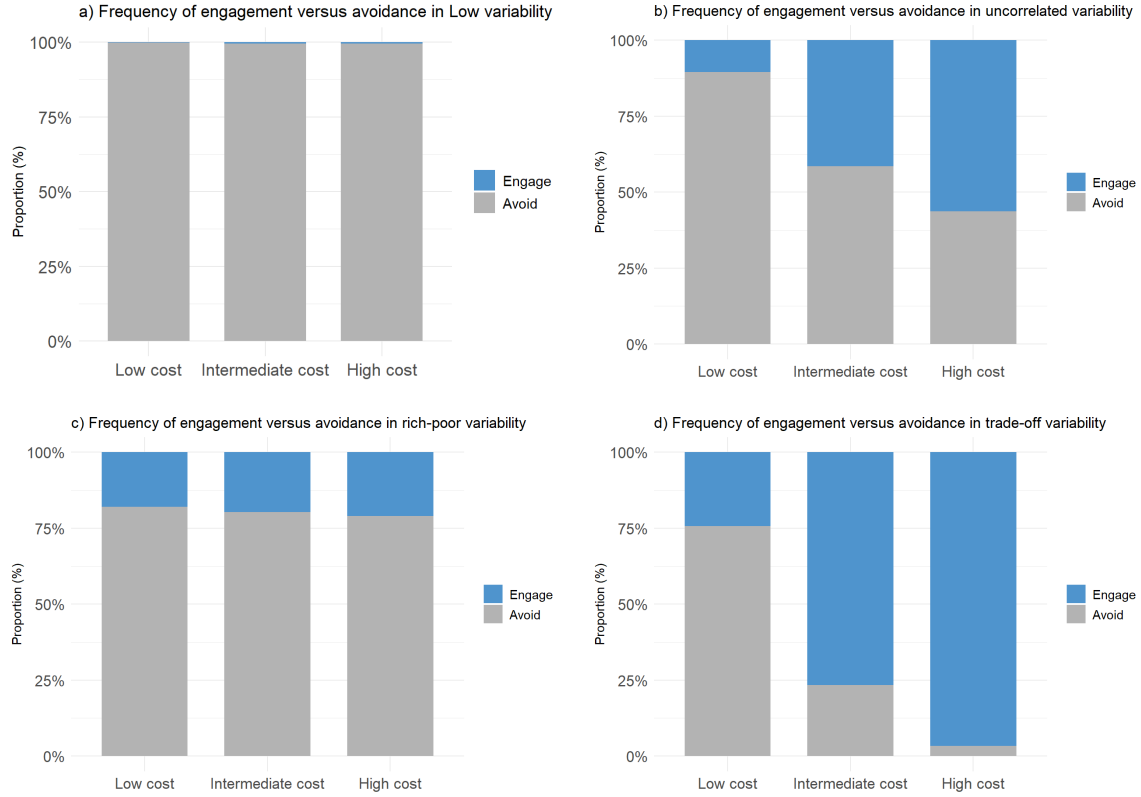


Figure S1: Action versus avoidance proportions across environmental variability scenarios. Panels (a–d) show the proportion of interactions in which agents engage in an action (blue) versus avoid interaction (grey), such that engagement and avoidance sum to 100% in each condition. Panel (a) corresponds to the low-variability environment, panel (b) to the uncorrelated variability environment, panel (c) to the rich–poor variability environment, and panel (d) to the trade-off variability environment. Within each panel, bars are shown separately for low, intermediate, and high conflict costs.

S1.2.1 Constrained avoidance

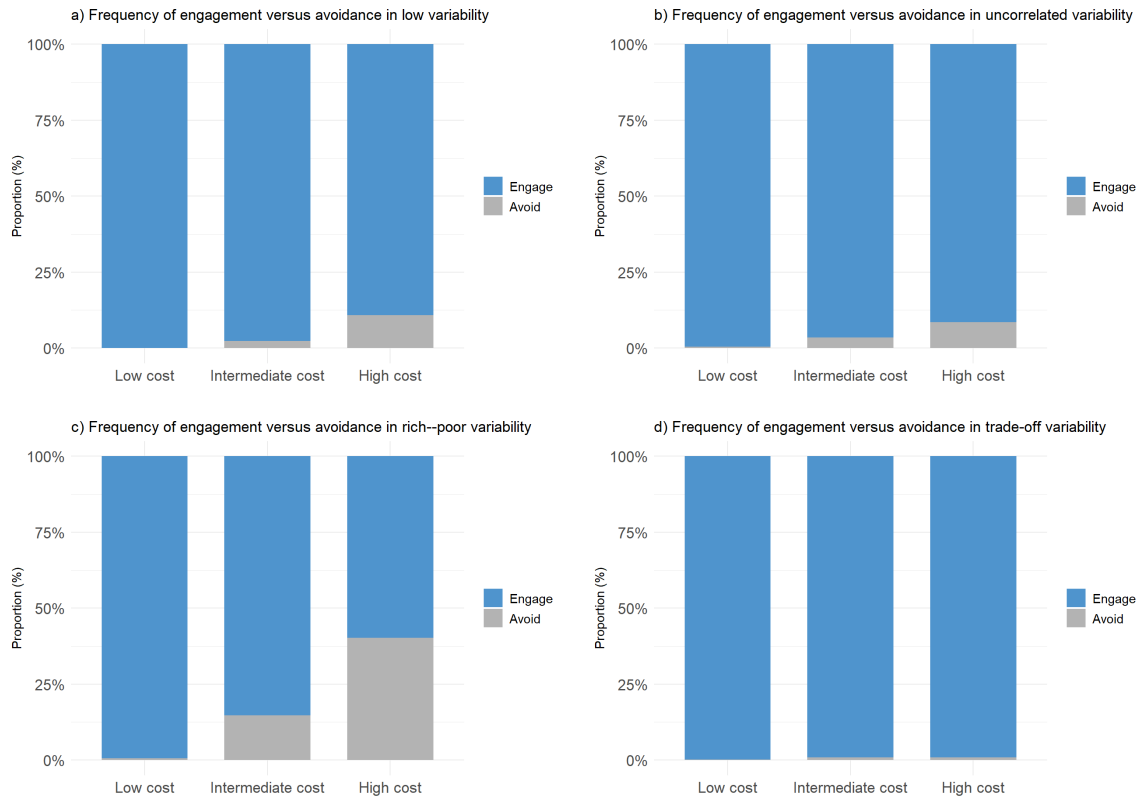


Figure S2: Action versus avoidance proportions across environmental variability scenarios when avoidance is constrained. Panels (a–d) show the proportion of interactions in which agents engage in an action (blue) versus avoid interaction (grey), such that engagement and avoidance sum to 100% in each condition. Panel (a) corresponds to the low-variability environment, panel (b) to the uncorrelated variability environment, panel (c) to the rich–poor variability environment, and panel (d) to the trade-off variability environment. Within each panel, bars are shown separately for low, intermediate, and high conflict costs.

S1.2.2 Complementary resources

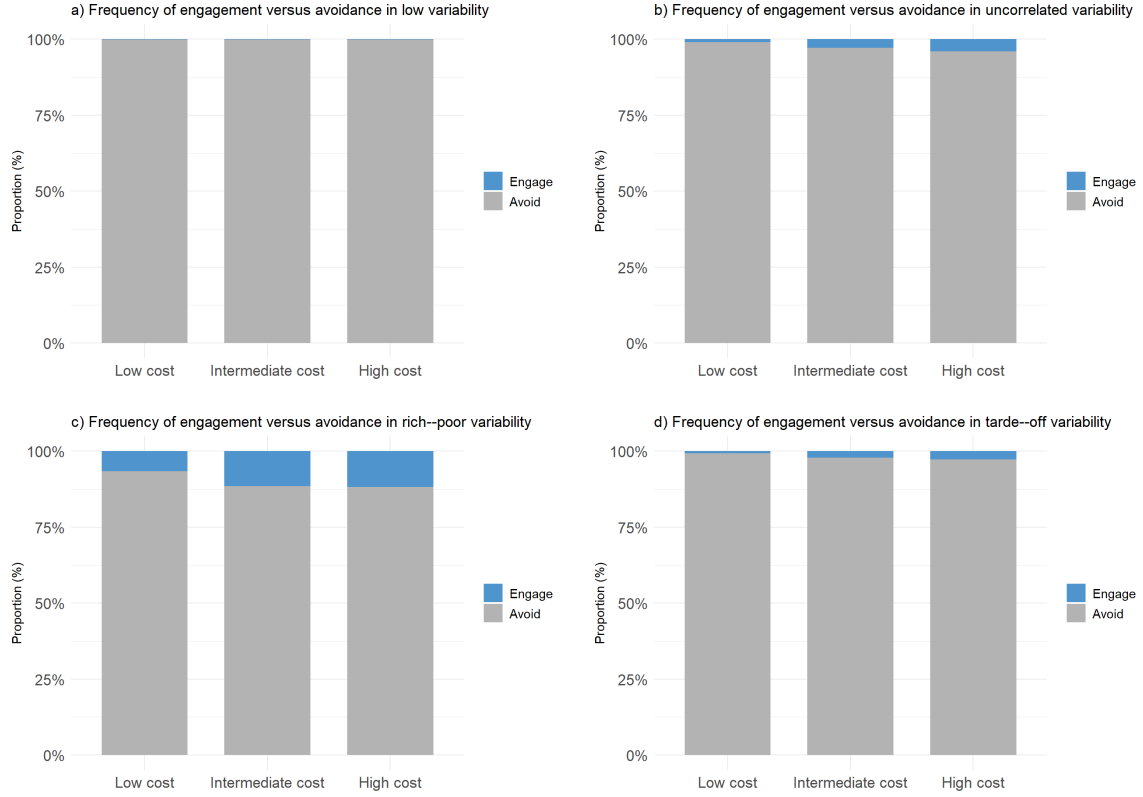


Figure S3: Action versus avoidance proportions across environmental variability scenarios under the Cobb–douglas formulation. Panels (a–d) show the proportion of interactions in which agents engage in an action (blue) versus avoid interaction (grey), such that engagement and avoidance sum to 100% in each condition. Panel (a) corresponds to the low-variability environment, panel (b) to the uncorrelated variability environment, panel (c) to the rich–poor variability environment, and panel (d) to the trade-off variability environment. Within each panel, bars are shown separately for low, intermediate, and high conflict costs.

S1.3 Within population variance

To clarify the relationship between within-population trait variation and the variation in actions reported in the main text, we provide additional analyses documenting the

distribution and covariance of evolving traits within populations. These figures complement the main results by making explicit whether behavioural heterogeneity arises primarily from differences among individuals or from shared reaction norms expressed under variable resource conditions.

Across all scenarios, equilibrium trait distributions were estimated by sampling 2,000 agents every 1,000 generations over the final 20,000 generations of each simulation, while population means and variances were computed using all $N = 10,000$ agents. Unless stated otherwise, all distributions and joint trait heatmaps therefore reflect pooled samples across time at stationarity.

S1.3.1 Low variability

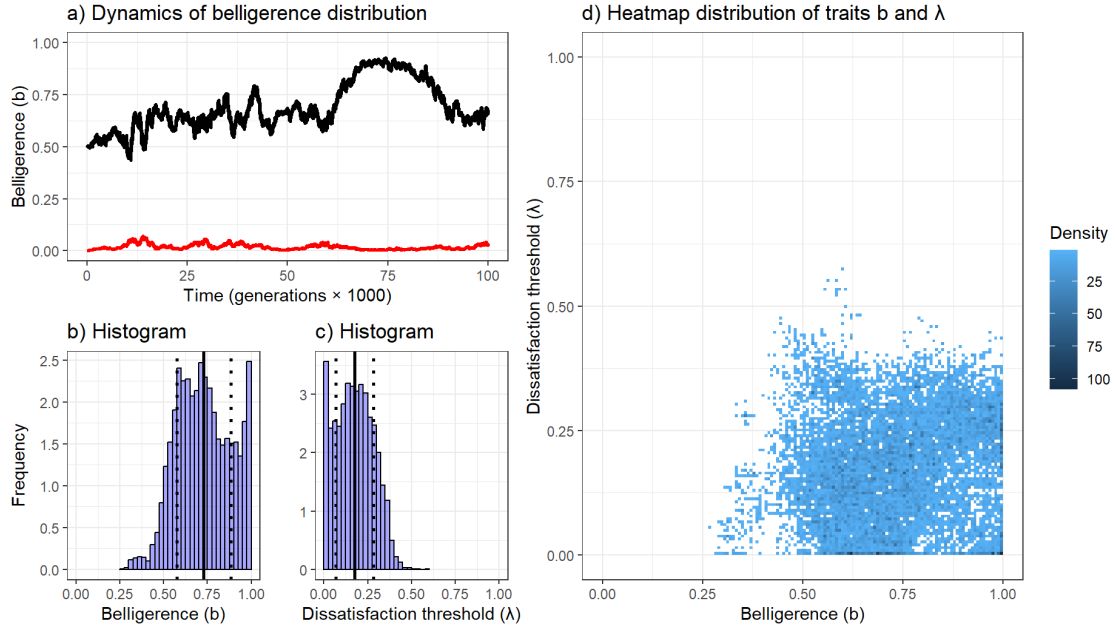


Figure S4: Trait dynamics and equilibrium distributions under the low-variability scenario and low conflict costs (1.25). Panel (a) shows the temporal dynamics of the population mean (black line) and within-population variance (red line) of belligerence (b). Panels (b) and (c) show the equilibrium distributions of belligerence (b) and dissatisfaction threshold (λ), respectively; solid black lines indicate population means and dotted lines indicate ± 1 SD. Panel (d) shows the joint equilibrium distribution of b and λ (heatmap).

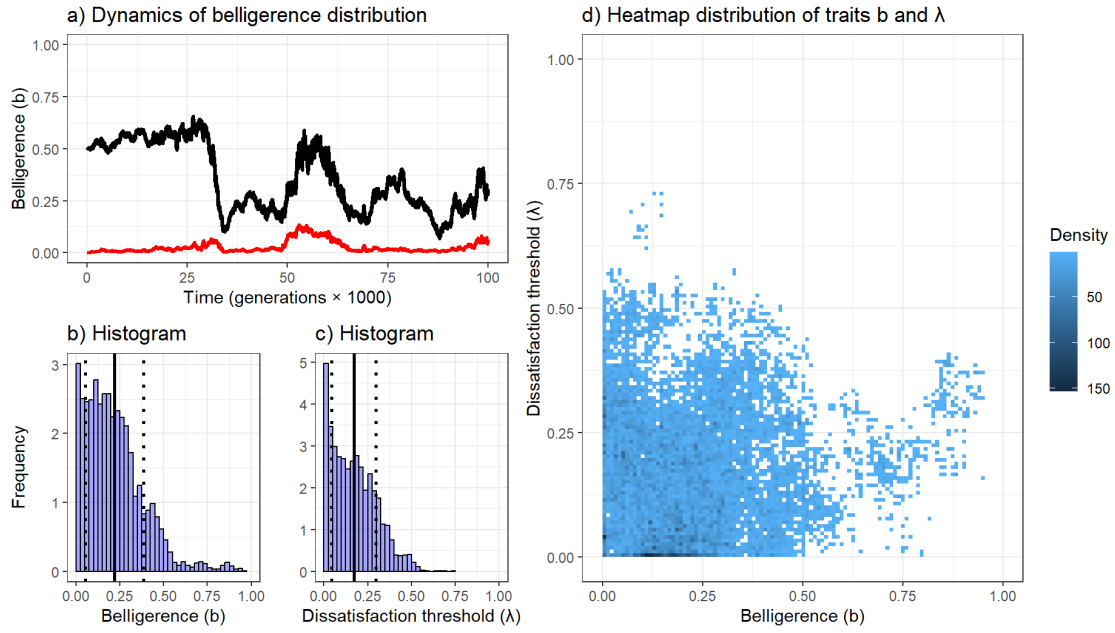


Figure S5: Trait dynamics and equilibrium distributions under the low-variability scenario and high conflict costs (7.0). Panel (a) shows the temporal dynamics of the population mean (black line) and within-population variance (red line) of belligerence (b). Panels (b) and (c) show the equilibrium distributions of belligerence (b) and dissatisfaction threshold (λ), respectively; solid black lines indicate population means and dotted lines indicate ± 1 SD. Panel (d) shows the joint equilibrium distribution of b and λ (heatmap).

S1.3.2 uncorrelated variability

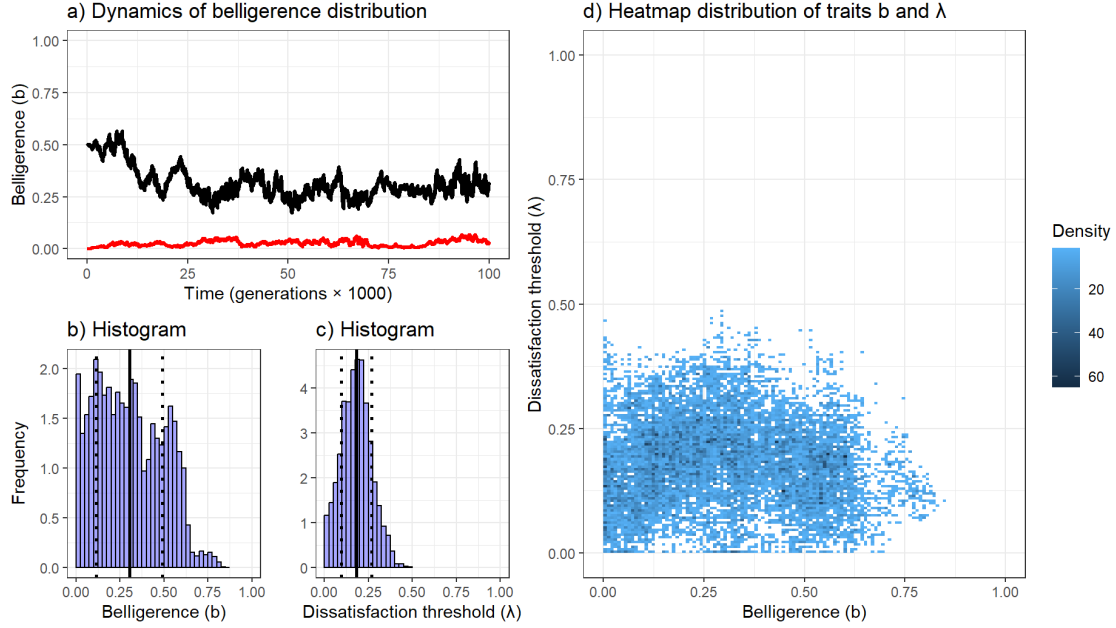


Figure S6: Trait dynamics and equilibrium distributions under the uncorrelated variability scenario and low conflict costs (1.25). Panel (a) shows the temporal dynamics of the population mean (black line) and within-population variance (red line) of belligerence (b). Panels (b) and (c) show the equilibrium distributions of belligerence (b) and dissatisfaction threshold (λ), respectively; solid black lines indicate population means and dotted lines indicate ± 1 SD. Panel (d) shows the joint equilibrium distribution of b and λ (heatmap).

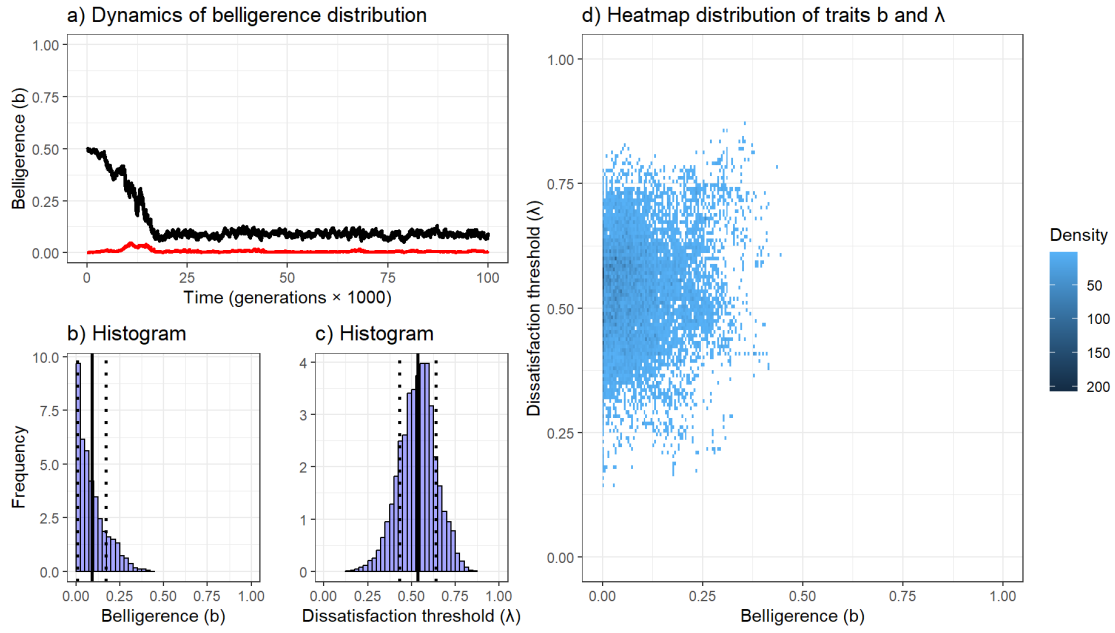


Figure S7: Trait dynamics and equilibrium distributions under the uncorrelated variability scenario and high conflict costs (7.0). Panel (a) shows the temporal dynamics of the population mean (black line) and within-population variance (red line) of belligerence (b). Panels (b) and (c) show the equilibrium distributions of belligerence (b) and dissatisfaction threshold (λ), respectively; solid black lines indicate population means and dotted lines indicate ± 1 SD. Panel (d) shows the joint equilibrium distribution of b and λ (heatmap).

S1.3.3 Rich–poor variability

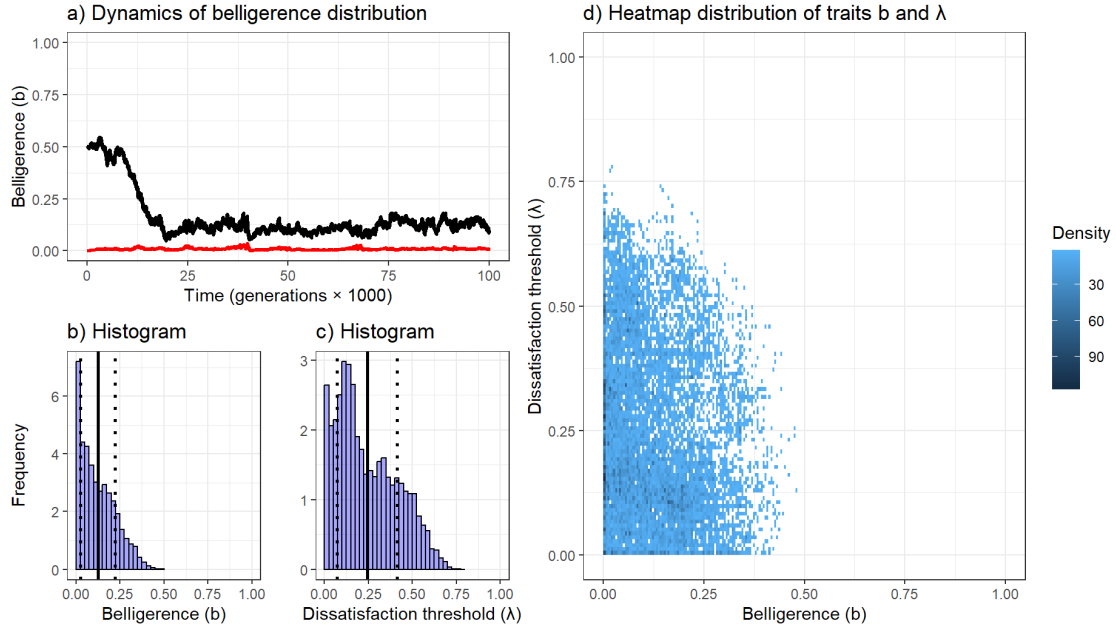


Figure S8: Trait dynamics and equilibrium distributions under the rich–poor variability scenario and low conflict costs (1.25). Panel (a) shows the temporal dynamics of the population mean (black line) and within-population variance (red line) of belligerence (b). Panels (b) and (c) show the equilibrium distributions of belligerence (b) and dissatisfaction threshold (λ), respectively; solid black lines indicate population means and dotted lines indicate ± 1 SD. Panel (d) shows the joint equilibrium distribution of b and λ (heatmap).

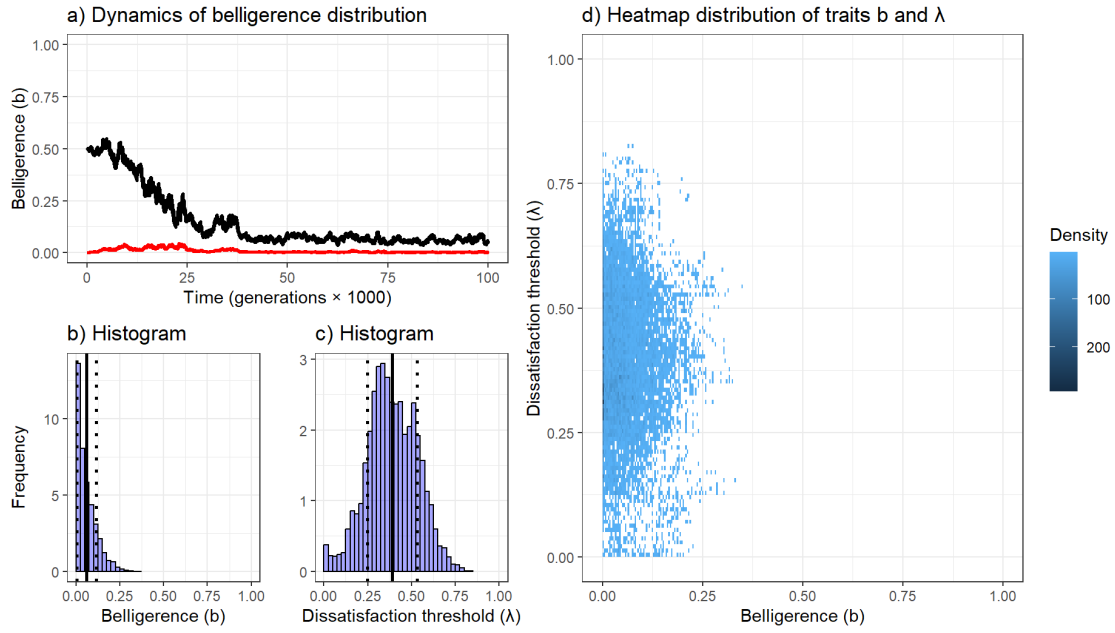


Figure S9: Trait dynamics and equilibrium distributions under the rich-poor variability scenario and high conflict costs (7.0). Panel (a) shows the temporal dynamics of the population mean (black line) and within-population variance (red line) of belligerence (b). Panels (b) and (c) show the equilibrium distributions of belligerence (b) and dissatisfaction threshold (λ), respectively; solid black lines indicate population means and dotted lines indicate ± 1 SD. Panel (d) shows the joint equilibrium distribution of b and λ (heatmap).

S1.3.4 Trade-off variability

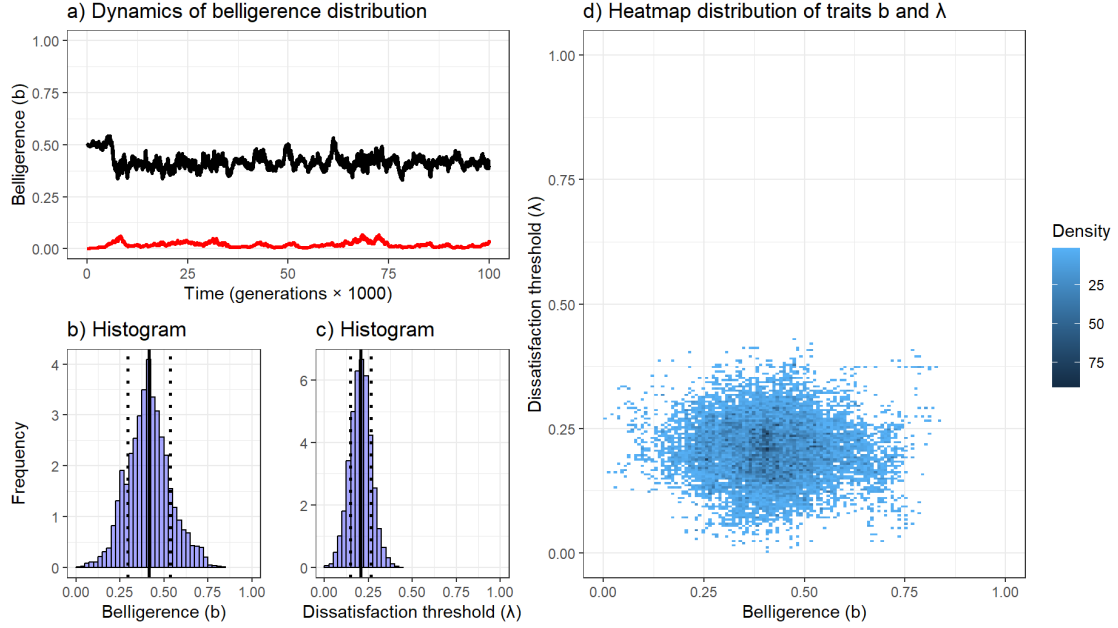


Figure S10: Trait dynamics and equilibrium distributions under the trade-off variability scenario and low conflict costs (1.25). Panel (a) shows the temporal dynamics of the population mean (black line) and within-population variance (red line) of belligerence (b). Panels (b) and (c) show the equilibrium distributions of belligerence (b) and dissatisfaction threshold (λ), respectively; solid black lines indicate population means and dotted lines indicate ± 1 SD. Panel (d) shows the joint equilibrium distribution of b and λ (heatmap).

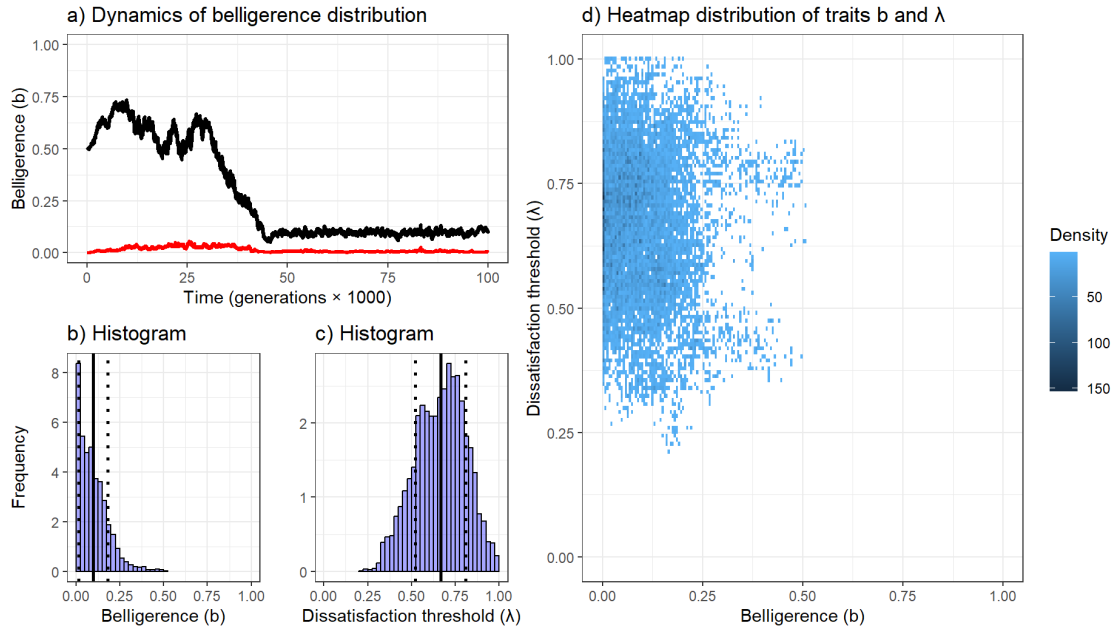


Figure S11: Trait dynamics and equilibrium distributions under the trade-off variability scenario and high conflict costs (7.0). Panel (a) shows the temporal dynamics of the population mean (black line) and within-population variance (red line) of belligerence (b). Panels (b) and (c) show the equilibrium distributions of belligerence (b) and dissatisfaction threshold (λ), respectively; solid black lines indicate population means and dotted lines indicate ± 1 SD. Panel (d) shows the joint equilibrium distribution of b and λ (heatmap).

S1.4 Constrained-avoidance case

In the constrained-avoidance case, avoidance no longer prevents exposure to conflict and is therefore treated as a peaceful interaction equivalent to trade; action proportions are thus computed over war, raid, and peaceful interactions (trade + avoidance). To be precise, When avoidance can no longer prevent exploitation by belligerent opponents, it becomes an inferior form of peaceful behaviour, it yields no advantage over trade against belligerent agents, while producing lower returns when paired with another non-belligerent agent. Under this rule, interactions reduce to three pairings, belligerent–belligerent, belligerent–non-belligerent, and non-belligerent–non-belligerent.

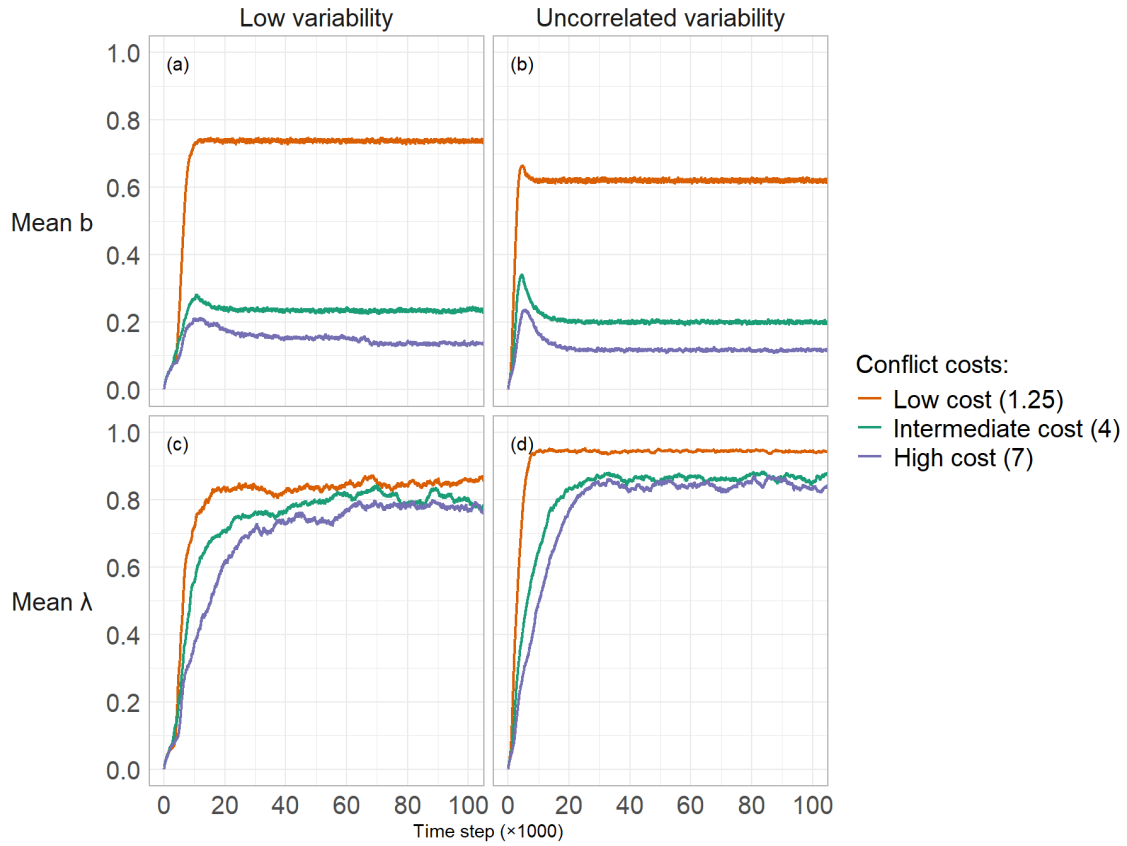


Figure S12: Time series of mean belligerence b and dissatisfaction threshold λ under low- and uncorrelated-variability environments in the constrained-avoidance case. Lines show means across 20 replicate simulations for different conflict-cost scenarios. Simulations run for 100,000 generations. Panels show (a) b under low variability, (b) b under uncorrelated variability, (c) λ under low variability, and (d) λ under uncorrelated variability.

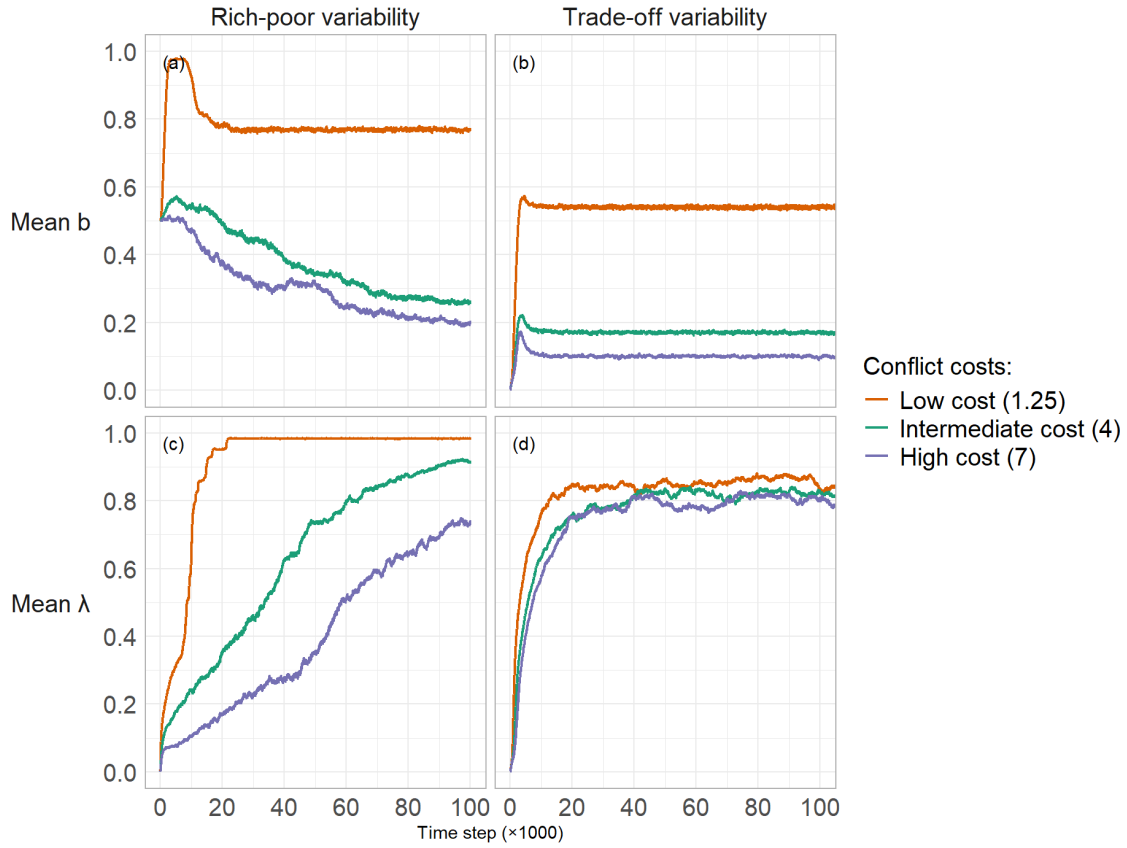


Figure S13: Time series of mean belligerence b and dissatisfaction threshold λ under the trade-off and rich-poor variability environments in the constrained-avoidance case. Lines show means across 20 replicate simulations for different conflict-cost scenarios. Simulations run for 100,000 generations. Panels show (a) b under low variability, (b) b under uncorrelated variability, (c) λ under low variability, and (d) λ under uncorrelated variability.

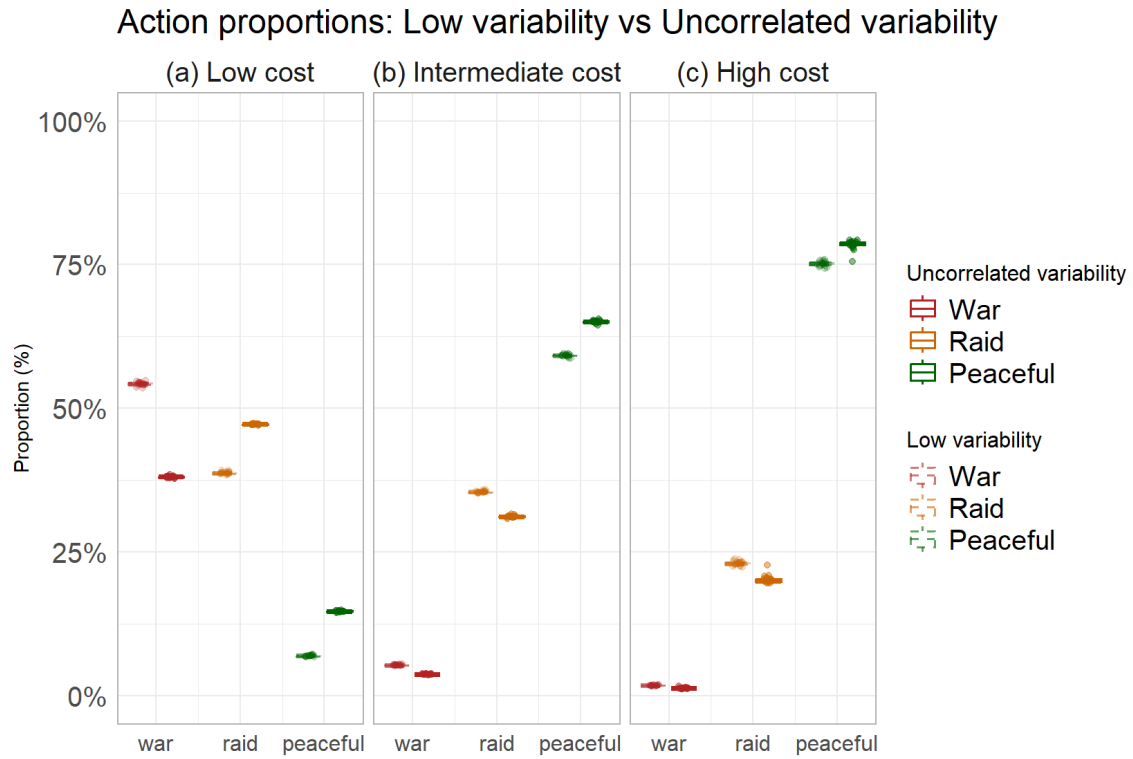


Figure S14: Action proportions among agents in low-variability and uncorrelated environments under (a) low (1.25), (b) intermediate (4), and (c) high (7) conflict costs in the constrained-avoidance case. Boxplots show the distribution of action frequencies across replicate simulations, with points indicating individual replicates. Proportions are calculated conditional on engagement, such that war, raid, and peaceful interactions (trade + avoidance) jointly sum to 100%. Values represent time-averaged proportions over the final 10,000 generations of each simulation (generations 190,000–200,000).

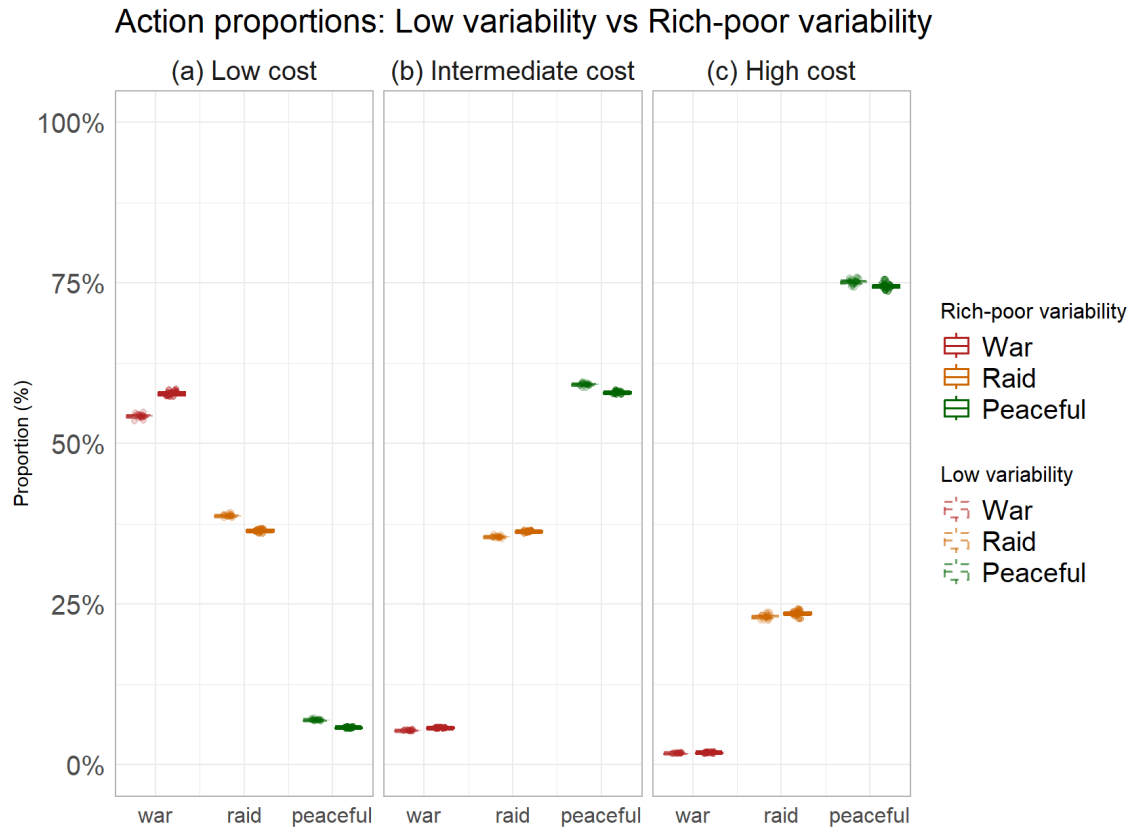


Figure S15: Action proportions among agents in low-variability and rich-poor environments under (a) low (1.25), (b) intermediate (4), and (c) high (7) conflict costs in the constrained-avoidance case. Boxplots show the distribution of action frequencies across replicate simulations, with points indicating individual replicates. Proportions are calculated conditional on engagement, such that war, raid, and peaceful interactions (trade + avoidance) jointly sum to 100%. Values represent time-averaged proportions over the final 10,000 generations of each simulation (generations 190,000–200,000).

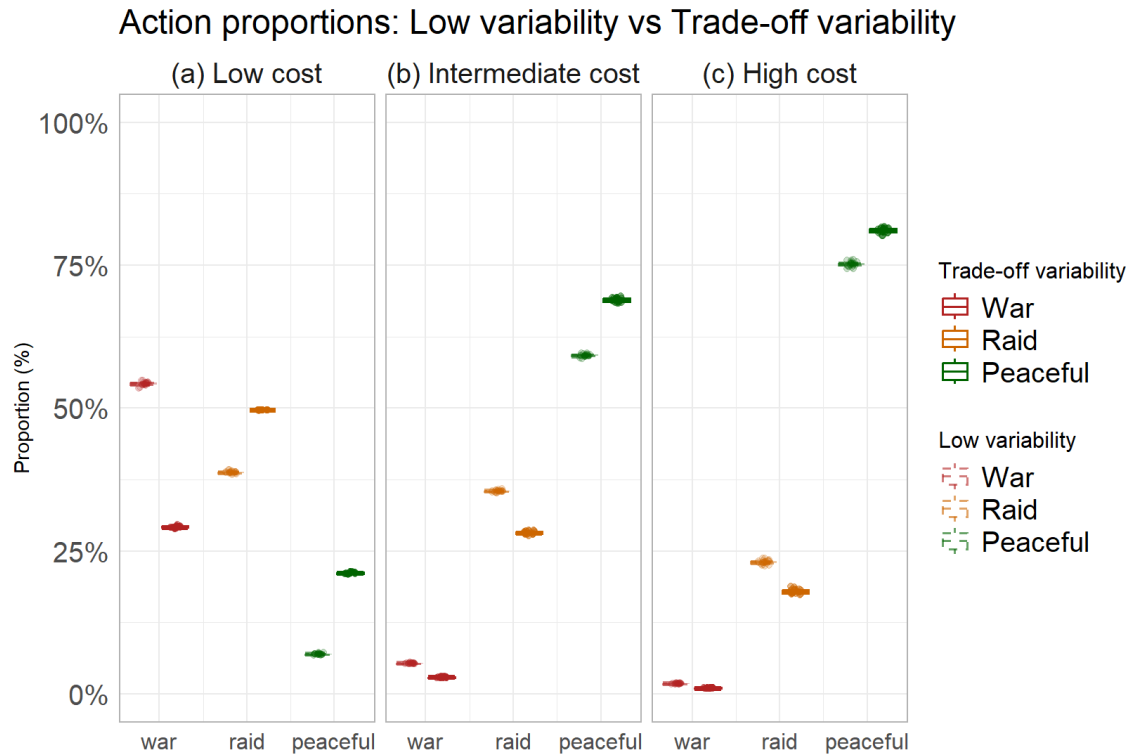


Figure S16: Action proportions among agents in low-variability and trade-off environments under (a) low (1.25), (b) intermediate (4), and (c) high (7) conflict costs in the constrained-avoidance case. Boxplots show the distribution of action frequencies across replicate simulations, with points indicating individual replicates. Proportions are calculated conditional on engagement, such that war, raid, and peaceful interactions (trade + avoidance) jointly sum to 100%. Values represent time-averaged proportions over the final 10,000 generations of each simulation (generations 190,000–200,000).

S1.5 Complementary resources

This section presents the analyses corresponding to the Cobb–Douglas formulation of fecundity described in section 3.2.5.

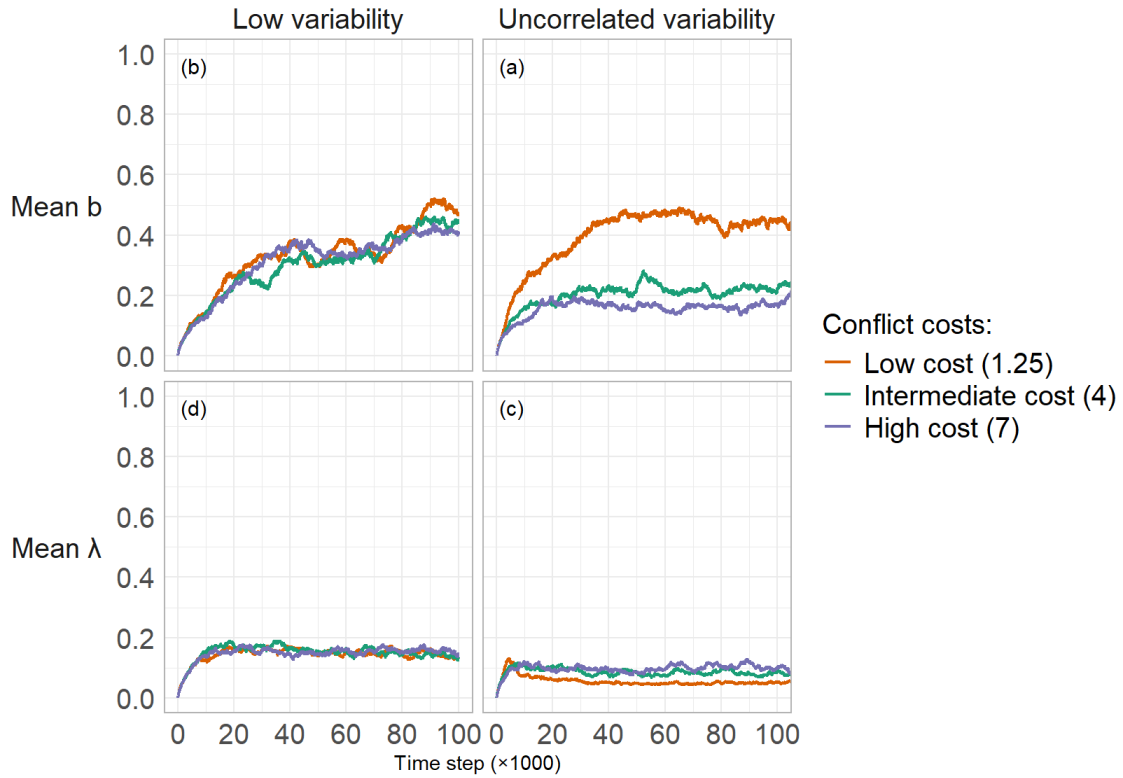


Figure S17: Time series of mean belligerence b and dissatisfaction threshold λ under low- and uncorrelated-variability environments in the Cobb–Douglas fecundity case. Lines show means across 20 replicate simulations for different conflict-cost scenarios. Simulations run for 100,000 generations. Panels show (a) b under low variability, (b) b under uncorrelated variability, (c) λ under low variability, and (d) λ under uncorrelated variability.

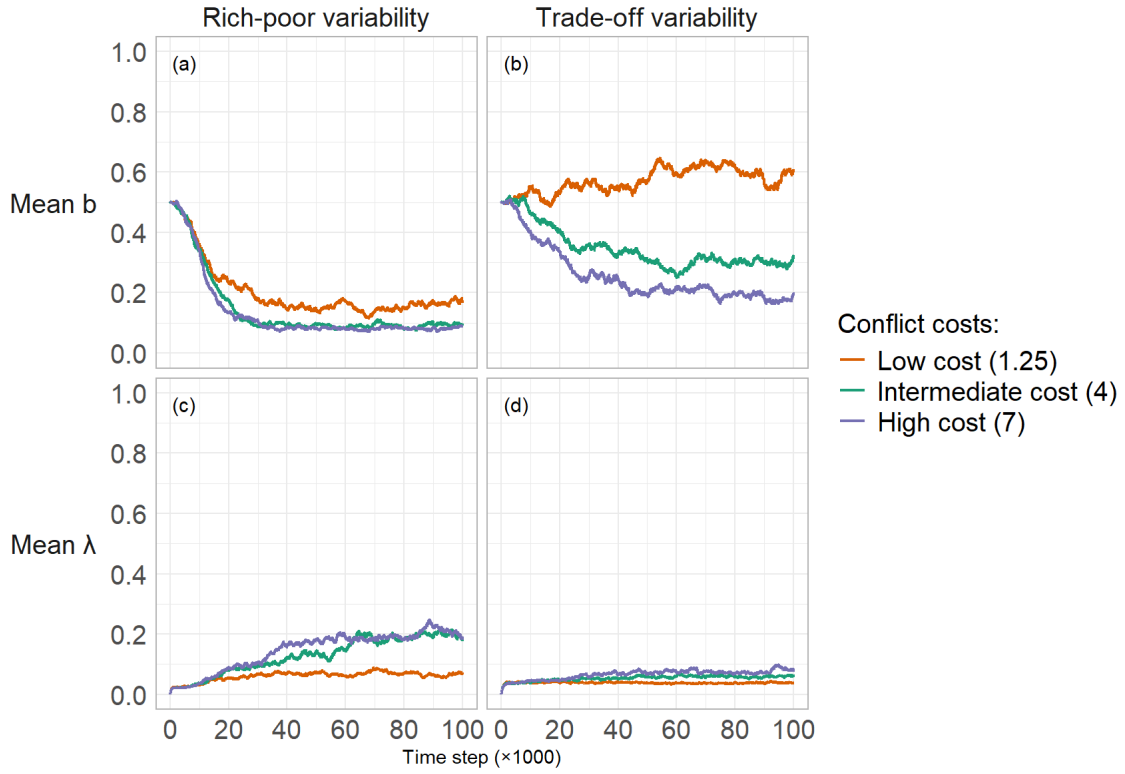


Figure S18: Time series of mean belligerence b and dissatisfaction threshold λ under the trade-off and rich-poor variability environments in the Cobb–Douglas fecundity case. Lines show means across 20 replicate simulations for different conflict-cost scenarios. Simulations run for 100,000 generations. Panels show (a) b under low variability, (b) b under uncorrelated variability, (c) λ under low variability, and (d) λ under uncorrelated variability.

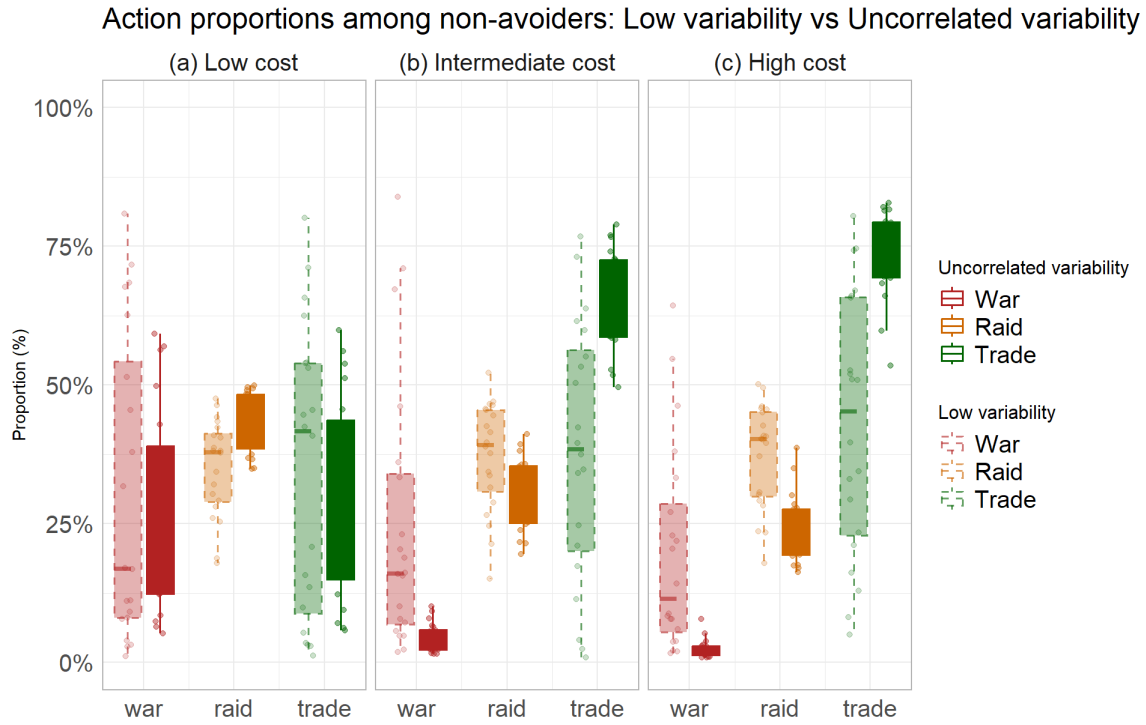


Figure S19: Action proportions among non-avoiding agents in the low-variability and uncorrelated environments under (a) low (1.25), (b) intermediate (4), and (c) high (7) conflict costs under the Cobb–Douglas fecundity formulation. For each cost scenario, boxplots show the distribution across replicate simulations and points indicate replicate values. Proportions are calculated conditional on engagement, such that raid, war, and trade jointly sum to 100%. Values correspond to time-averaged proportions computed over the final 10,000 generations of each simulation (generations 190,000–200,000).

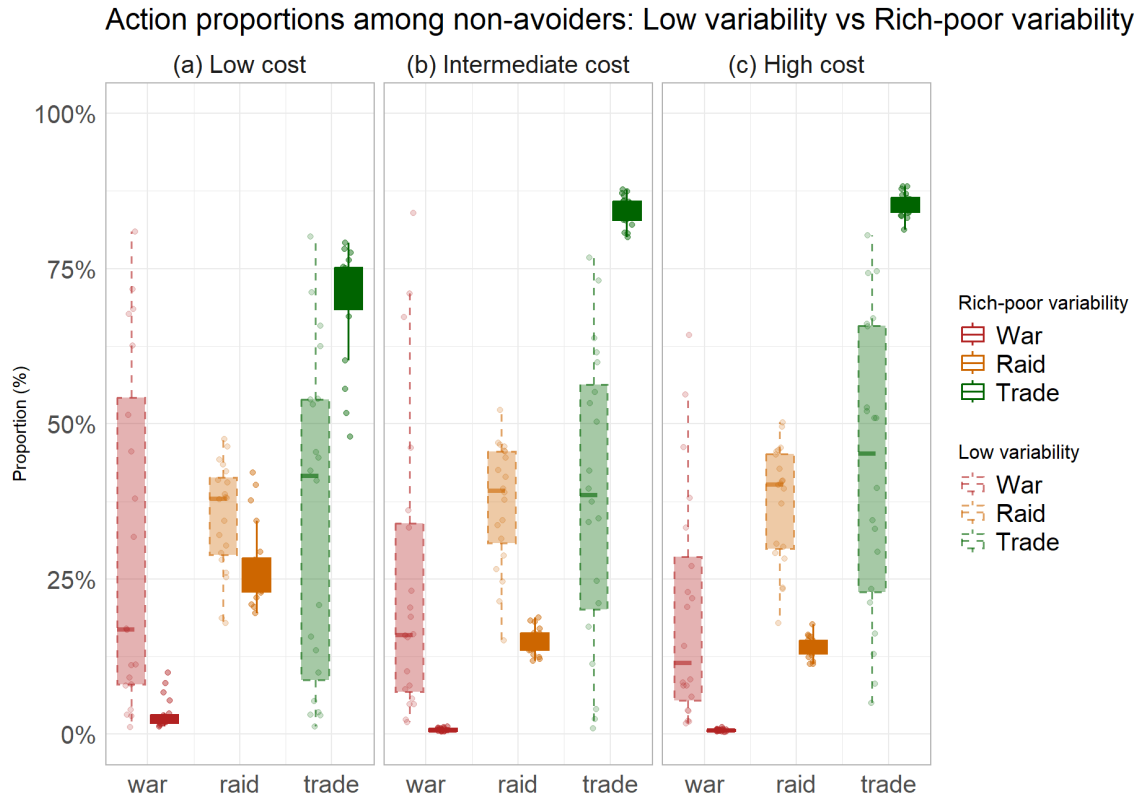


Figure S20: Action proportions among non-avoiding agents in the low-variability and rich-poor environments under (a) low (1.25), (b) intermediate (4), and (c) high (7) conflict costs under the Cobb–Douglas fecundity formulation. For each cost scenario, boxplots show the distribution across replicate simulations and points indicate replicate values. Proportions are calculated conditional on engagement, such that raid, war, and trade jointly sum to 100%. Values correspond to time-averaged proportions computed over the final 10,000 generations of each simulation (generations 190,000–200,000).

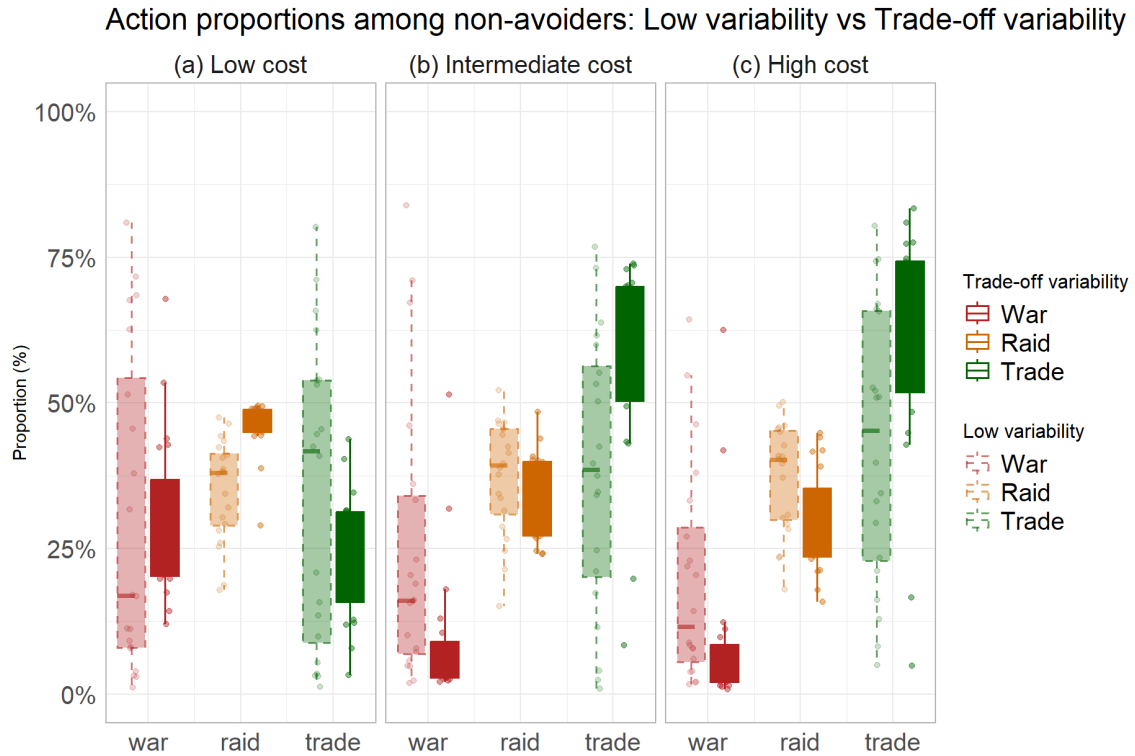


Figure S21: Action proportions among non-avoiding agents in the low-variability and trade-off environments under (a) low (1.25), (b) intermediate (4), and (c) high (7) conflict costs under the Cobb–Douglas fecundity formulation. For each cost scenario, boxplots show the distribution across replicate simulations and points indicate replicate values. Proportions are calculated conditional on engagement, such that raid, war, and trade jointly sum to 100%. Values correspond to time-averaged proportions computed over the final 10,000 generations of each simulation (generations 190,000–200,000).